END 308 – Facilities Planning and Design
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PROJECT

Location Optimization and Layout Analysis for TotalEnergies Logistics Network in Berlin

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ABSTRACT

This project presents an integrated method for identifying the optimal location and internal layout of a new TotalEnergies service station in Berlin. First, a Single Facility Rectilinear Location Problem (SFRLP) is formulated to minimize the weighted distance to existing stations. District-specific weights—based on population, traffic volume, and industrial activity—are derived using the Analytic Hierarchy Process (AHP). After linearizing the model and solving it in Python/Pyomo, the initial optimal point in Friedrichshain-Kreuzberg was found to be legally infeasible. A circular forbidden-zone constraint (11 km radius) was introduced, and iterative re-optimization yielded a feasible and operationally favorable location in Tempelhof-Schöneberg.

In the second phase, internal layout planning is conducted using REL Charts, ALDEP, and distance-based evaluation. Four layouts were generated, and the best one was refined with the CRAFT heuristic, resulting in a significant reduction in internal movement cost. The study provides a practical foundation for future facility planning.

Keywords: Facility Location, SFRLP, Facility Layout, Optimization, Berlin.

1. Introduction

The main goal of this project is to find the best location for a new TotalEnergies service station in Berlin, Germany. Choosing the right location is very important for reducing costs and reaching more customers. It will provide not only traditional fuel but also electric vehicle (EV) charging and a convenience store.

In the first part of the study, The main focus was finding the optimal coordinates (x*, y*) for this single facility. To do this scientifically,a mathematical model was used called the Single Facility Rectilinear Location Problem.The Rectilinear (L1) distance metric (Manhattan distance) was chosen because it represents city driving better than straight-line distance. Also, to make the model realistic, Analytic Hierarchy Process (AHP) was used to calculate "demand weights." This allowed them to prioritize areas with high traffic, population, and industrial activity, rather than treating every location the same.

The second part of this report focuses on the Facility Layout Design. In that section, the internal arrangement of fuel pumps, the store, and other services is determined using logic from the ALDEP and CRAFT algorithms to ensure safe and smooth traffic flow.

2. Methods

This section outlines the mathematical models, data collection strategies, and cost analysis frameworks employed to solve the facility location and layout problems.

2.1. Data Collection Strategy

In order to ensure that the facility location decision was based on empirical evidence and reliable urban data, several quantitative location criteria were selected and evaluated. These criteria reflect key demographic, transportation, and economic characteristics of the study area. The evaluated criteria are presented below. The official data sources from which they were obtained are provided in the References section.

- Population Demographics: District-level population data were used to identify high-density residential zones with greater potential fuel demand [1].
- Vehicle Ownership: Motorization rates were used as a proxy for private vehicle usage intensity across districts [2].
- Traffic Volume (DTV): Daily traffic flow data on major arterial roads were used to detect high-exposure transportation corridors [3].
- Industrial Zones: Land-use data were analyzed to incorporate the effect of commercial and industrial fuel demand [4].
- Existing Network: The coordinates of existing TotalEnergies stations were used to account for the current network structure and competitive spacing [5].

A comprehensive summary of the specific data values derived from online sources for each administrative district is presented in Table 1.

Table 1. Key Demographic, Traffic, and Industrial Indicators for Berlin Districts.

District	Population [1]	Vehicle Ownership (Cars/1000 Inh.) [2]	Critical Road & Daily Traffic Volume [3]	Industrial Area (Hectares) [4]
Mitte	397,134	231	B1 / Leipziger Str. (~55,000)	185 ha
F-Kreuzberg	293,454	204	B96 / Mehringdamm (~45,000)	120 ha
Pankow	424,307 (Highest)	315	A114 / Prenzlauer Prom. (~60,000)	210 ha
Charlottenburg	343,081	362	A100 (City Ring) (~165,000)	255 ha
Spandau	257,091	421	B2 / Heerstraße (~48,000)	1,150 ha
Steglitz-Zehl.	310,071	455	A103 / Schloßstr. (~50,000)	190 ha
Tempelhof-Sch.	355,868	368	A100 / A103 Interchange (~140,000)	780 ha (Industrial Zone)
Neukölln	330,017	285	A100 / Britzer Damm (~110,000)	410 ha
Treptow-Köp.	294,081	415	B96a / Adlergestell (~65,000)	980 ha
Marzahn-Hell.	291,948	395	B1 / Alt-Biesdorf (~70,000)	850 ha
Lichtenberg	311,881	328	B1 / Frankfurter Allee (~60,000)	520 ha
Reinickendorf	268,792	430	A111 (Hamburg Route) (~90,000)	640 ha

2.2. Methods and Model

The problem has been discussed in several types of well known facility models such as Single Facility Models, Multiple Facility Models,Capacitated Facility Location Model,Uncapacitated Facility Location Model, P-Center or P-Median models. It is decided that the gas station vision must minimize the total distance instead of minimizing maximum distance.The underlying philosophy behind the gas stations is not to be very close to certain specific customers and far from others, but to be close to everyone. Therefore, the total distance has been minimized. Total distance is calculated with respect to weights of fuel stations between an existing gas station and a newly established gas station pair.The special weight was formulated by using multiple factors and criteria such as population of the district, the traffic density of the district, among others. The distance type which is used in objective function also be selected between euclidean distance,squared euclidean distance chebyshev distance and rectilinear (Manhattan) distance.Since the problem contains gas stations it is also related the other factors such as population cars and especially road network and traffic flow. Rectilinear distance was selected in order to calculate the distances between gas stations in a way that is compatible with actual vehicle movement and to incorporate them into the model.Also instead of multiple-facility model, a single-facility model was used, as the goal of the company "TotalEnergies" is to construct one gas station in Berlin as a quality rather than quantity vision .Since the aim is to establish more efficient and profitable stations compared to existing ones,The improvement methods has been used such as AHP method to calculate inner weight which is used in model. The model was improved based on the pure single-facility rectilinear distance model.

- Objective: Minimize total weighted distance between existing ones and newly opened ones .
- Additional Constraint: A Euclidean distance of at least 11 km is imposed between any pair of new facilities to prevent cannibalization .
- Additional Property: Iso-cost lines are constructed to analyse the location more properly and decide any other locations which are suitable.

Non-Linear Single Facility Rectilinear Model

The model determines the optimal geographic position of a new facility so as to minimize the total interaction cost between new facilities and existing ones, as well as the interaction cost among the new facilities themselves.

I. $Sets$

$I = \{1, 2,..., m\} : set\ of\ existing\ facilities$

II. $Parameters$

$a_i\ (i \in I) : Longitude\ of\ the\ existing\ facilities.$

$b_i\ (i \in I) : Latitude\ of\ the\ existing\ facilities.$

$w_i\ (i \in I) : Weight(importance\ score)\ of\ existing\ facility\ i.$

III. $Decision\ Variables$

$x \in R : Longitude\ of\ the\ new\ facility$

$y \in R$: $Latitude\ of\ the\ new\ facility$

IV. $Objective\ Function\ And\ Model$

$$min\ Z = \sum_{i \in I} w_i \left| x - P_i \right| + \sum_{i \in I} w_i \left| y - P_i \right|$$

st:

x : urs

y : urs

Linear Single Facility Rectilinear Model

Since a pure single facility rectilinear distance model is not a linear model due to the absolute function on the objective function it must be linearized. To linearize a single facility rectilinear distance model 4 additional variables have been defined to calculate the amount of positive and negative deviation with respect to. 0. This approach is the result of the property of absolute value function. Due to new variables additional constraints are also inserted into the model.

I. $Sets$

$I = \{1, 2, ..., m\}$: $set\ of\ existing\ facilities$

II. $Parameters$

$a_i\ (i \in I)$: $Longitude\ of\ the\ existing\ facilities.$

$b_i\ (i \in I)$: $Latitude\ of\ the\ existing\ facilities.$

$w_i\ (i \in I)$: $Weight(importance\ score)\ of\ existing\ facility\ i.$

III. $Decision\ Variables$

$x \in R$: $Longitude\ of\ the\ new\ facility$

$y \in R$: $Latitude\ of\ the\ new\ facility$

r_i, s_i, u_i, t_i: $i \in I$ $Linearization\ variables$

IV. $Objective\ Function\ And\ Model$

$$min\ Z = \sum_{i \in I} w_i(r_i + s_i + u_i + t_i)$$

st:

$x - a_i = r_i - s_i\ \forall\ i \in I$

$y - b_i = u_i - t_i\ \forall\ i \in I$

$r_i, s_i, u_i, t_i \geq 0\ \forall\ i \in I$

x : urs

y : urs

Weight Calculation for The Model

To improve the performance of the single-facility rectilinear location model a more accurate weighting structure was required in order to evaluate the demand points. For this purpose, Analytic Hierarchy Process (AHP) was implemented to calculate inner weights of the criteria.

```
A = np.array([
    # Population  Traffic Industry
    [1.00,  1/3,  0.66  ],  # Population
    [3.00,  1.00,  2.00  ],  # Traffic
    [1.5,   1/2,  1.00  ]  # Industry
])
```

Figure 1. Objective Comparison Matrix

First a pairwise comparison matrix A was constructed as shown in Equation (1), where each entry a_{ij} represents the relative importance of criterion i compared to criterion j:

$$A = [a_{ij}] \quad i, j \in \{1, 2, 3\} \quad (1)$$

Next in order to AHP methodology, each column of the comparison matrix has been normalized.This is achieved by dividing each element a_{ij} by the sum of its corresponding column, denoted by s_j in Equation (2):

$$s_j = \sum_{i=1}^{3} a_{ij} \quad j \in \{1, 2, 3\} \quad (2)$$

Using the column sums, the normalized comparison matrix A^{norm} is obtained as given in Equation (3):

$$A^{norm} = [a_{ij}^{norm}] \quad i, j \in \{1, 2, 3\} \quad (3)$$

Each normalized element is computed by:

$$a_{ij}^{norm} = \frac{a_{ij}}{s_j} \quad i, j \in \{1, 2, 3\} \quad (4)$$

After the normalization step, the inner weights of the criteria is calculated by averaging each row of the normalized matrix as shown in Equation (5):

$$wa_i = \frac{1}{3} \sum_{j=1}^{3} a_{ij}^{norm} \quad \forall i \in \{1, 2, 3\} \quad (5)$$

Finally, the resulting of weights is expressed as:

$$Alternatives\ Weights = [wa_1, wa_2, wa_3]_{1x3}$$

These weights are then integrated into the rectilinear distance minimization model to ensure that the relative importance of each criterion is accurately reflected in the final objective function.

Population density , traffic density and industry zone scores are also normalized in order to reach a more proper and balanced weight value.

$W_i = wi_1 x\, population\ density\ + wi_2\, x\, traffic\ density\ +\ wi_3\, x\, industry\ zone\ score$

```
# AHP weights and Zone weights
df["weight"] = (
    weights[0] * df["Nüfus (1-10)"] +
    weights[1] * df["Trafik & Araç (1-10)"] +
    weights[2] * df["Sanayi (1-10)"]
)
df["weight"] = df["weight"].fillna(df["weight"].mean())

# Temiz indeks
df.reset_index(drop=True, inplace=True)
```

Figure 2. Weight Calculation

Improved Linear Single Facility Rectilinear Model

Single Facility Rectilinear Model is a suitable approach for the vision of the problem, as its inputs and parameters fit well with the data which is gathered from Kaggle and various resources.But still there are some incompatibilities. Based on the collected data and the analysis of population distribution in Berlin, it is said that the optimal location which is located by solving the model may not be suitable to construct a gas station because of environmental and governmental problems. To overcome this issue, a circular forbidden zone with a radius of R km has been created around the optimal location obtained from the initial solution.The model is repeatedly re-solved until a feasible location outside the forbidden zone has obtained and an additional constraint has been added to the model to ensure that the new optimal point does not fall within this circular area.   This condition has been inserted into model as an additional constraint and distance between found optimal solution and new optimal solution has been calculated by using euclidean distance instead of rectilinear distance euclidean distance has been preferred because this measurement is treated not as a rigid,fixed distance but as a smoother and more continuous representation.

Linear Single Facility Rectilinear Model for Initial Solution

I. *Sets*

$I\ = \{1, 2,..., m\}:\ set\ of\ existing\ facilities$

II. $Parameters$

$a_i\ \ (i\ \in I):\ Longitude\ of\ the\ existing\ facilities.$

$b_i\ \ (i\ \in I):\ Latitude\ of\ the\ existing\ facilities.$

$w_i\ \ (i \in I):\ Weight(importance\ score)\ of\ \ existing\ facility\ i.$

III. $Decision\ Variables$

$x \in R$: $Longitude\ of\ the\ new\ facility$

$y \in R$: $Latitude\ of\ the\ new\ facility$

$r_{ji}, s_{ji}, u_{ji}, t_{ji}$: $(j, i) \in I_2$:

$Linearization\ variables\ for\ distances\ between\ new\ and\ existing\ ones$

IV. $Objective\ Function\ And\ Model\ for\ Initial\ Solution$

For the initial solution, a pure single facility rectilinear LP model has been solved through Gurobi solver.

$$min\ Z = \sum_{i \in I} w_i(r_i + s_i + u_i + t_i)$$

st:

$x - a_i = r_i - s_i \ \forall\, i \in I$ (1)

$y - b_i = u_i - t_i \ \forall\, i \in I$ (2)

$r_i, s_i, u_i, t_i \geq 0 \ \forall\, i \in I$

x : urs

y : urs

Linear Single Facility Rectilinear Model for Advanced Solution

After the initial solution model is iteratively solved until a valid solution can be seen. For each iteration the third constraint which is shown below is inserted to satisfy forbidden area property.

I. $Sets$

$I = \{1, 2, ..., m\}$: $set\ of\ existing\ facilities$

II. $Parameters$

a_i $(i \in I)$: $Longitude\ of\ the\ existing\ facilities.$

b_i $(i \in I)$: $Latitude\ of\ the\ existing\ facilities.$

w_i $(i \in I)$: $Weight(importance\ score)\ of\ existing\ facility\ i.$

$Rad \in R$: $radius\ of\ the\ forbidden\ zone.$

x_1 : $Longitude\ solution\ of\ step 1.$

x_2 : $Longitude\ solution\ of\ step 2.$

b_i $(i \in I)$: $Latitude\ of\ the\ existing\ facilities.$

III. $Decision\ Variables$

$x \in R$: $Longitude\ of\ the\ new\ facility$

$y \in R$: $Latitude\ of\ the\ new\ facility$

$r_{ji}, s_{ji}, u_{ji}, t_{ji}$: $(j, i) \in I_2$:

$Linearization\ variables\ for\ distances\ between\ new\ and\ existing\ ones$

IV. $Objective\ Function\ And\ Model\ for\ Advanced\ Solution$

$$min\ Z = \sum_{i \in I} w_i (r_i + s_i + u_i + t_i)$$

st:

$x - a_i = r_i - s_i \ \forall\, i \in I$ (1)

$y - b_i = u_i - t_i \ \forall\, i \in I$ (2)

Constraint (3) ensures that the new facility is not placed inside the forbidden circular zone. The point (x_0, y_0) is the center of the restricted area, and the radius *Rad* (in kilometers) is converted into degrees by dividing by 111. Constraint (3) forces the solution (x, y) to stay outside this circle, thereby satisfying legal or environmental restrictions.

$(x - x_0)^2 - (y - y_0)^2 \geq (\frac{Rad}{111})^2$ (3)

$r_i, s_i, u_i, t_i \geq 0 \ \forall\, i \in I$

$x : urs$

$y : urs$

Python Implementation of Linear Single Facility Rectilinear Model Constraints for Advanced Solution

```
#Parse the coordinates if there is a typo
def parse_lat_lon_raw(val):
    try:
        f_val = float(val)
        if f_val > 1000:  # convert  52123456 ---->  52.123456
            return f_val / 1_000_000
        return f_val
    except:
        return None
```

Figure 3. Parse Coordinates Function

In case of any typographical inconsistencies in the latitude or longitude values, the coordinates were parsed and corrected to ensure a consistent and valid format.

```
# Sets Exist Stations
I = list(df.index)
Model.I = Set(initialize=I)

# Parameteres: Coordinates and Weights
a_dict = {i: df.loc[i, "longitude"] for i in I}
b_dict = {i: df.loc[i, "latitude"]  for i in I}
w_dict = {i: df.loc[i, "weight"]    for i in I}

Model.a = Param(Model.I, initialize=a_dict)  # lon
Model.b = Param(Model.I, initialize=b_dict)  # lat
Model.w = Param(Model.I, initialize=w_dict)  # weight

# Decision var:  New Facility Coordinates

# x and y variable
Model.X = Var(within=Reals)
Model.Y = Var(within=Reals)

# Linearization variables
Model.r = Var(Model.I, within=NonNegativeReals)
Model.s = Var(Model.I, within=NonNegativeReals)
Model.u = Var(Model.I, within=NonNegativeReals)
Model.t = Var(Model.I, within=NonNegativeReals)
```

Figure 4. Sets, Parameters, Decision Variables

The set I has been defined.Parameters a_i, b_i, w_i are defined and they are defined as a dictionary list structure to make program easy to access and more dynamic.After that variables x and y are defined finally to linearize the objective function linearization variables $r_i s_i, u_i, t_i$ are defined.

```
# (a) Rectilinear Distance Linearization
def x_diff_rule(M, i):
    # X - a_i = r_i - s_i
    return M.X - M.a[i] == M.r[i] - M.s[i]
Model.CX = Constraint(Model.I, rule=x_diff_rule)

def y_diff_rule(M, i):
    # Y - b_i = u_i - t_i
    return M.Y - M.b[i] == M.u[i] - M.t[i]
Model.CY = Constraint(Model.I, rule=y_diff_rule)
```

Figure 5. Rectilinear Distance Linearization

The rectilinear distance linearization constraints have been incorporated into the model to ensure that the absolute distance terms are represented in a linear and solver-compatible form.

```
# (b) FORBIDDEN ZONE CONSTRAINT: CREATE A FORBIDDEN ZONE R km radius circle center coordinates are x0 y0
def forbidden_zone_rule(M):
    return (M.X - FORBID_LON)**2 + (M.Y - FORBID_LAT)**2 >= FORBID_DEG2
Model.ForbiddenZone = Constraint(rule=forbidden_zone_rule)
```

Figure 6. Forbidden Zone Constraint

Following the initial solution, the forbidden zone constraint was introduced into the subsequent model iterations. This constraint utilizes the x and y coordinates of the initially identified optimal point to construct a restricted region in which no new facility can be located. When this constraint is omitted, the model reduces back to its original form used in the initial solution phase.

```
# ==============================
# 5) Objective Function
# ==============================
# Minimize Σ_i w_i (|X - a_i| + |Y - b_i|)
def obj_rule(M):
    return sum(
        M.w[i] * (M.r[i] + M.s[i] + M.u[i] + M.t[i])
        for i in M.I
    )
Model.Obj = Objective(rule=obj_rule, sense=minimize)
```

Figure 7. Objective Function

The objective function has been constructed by using the rules of the single-facility rectilinear model.

3. Results, Outputs and Alternative Solutions of The Facility Location

3.1. Initial Solution

In order to visualize the outcome of the initial model, the rectilinear location model was solved using the existing facility data.The map below presents the spatial distribution of the current facilities together with the optimal facility location estimated by the model.

According to the initial solution, the model identified the following coordinates as the optimal location for the new facility:

- Latitude:52.518378
- Longitude:13.440946
- Objective Function:32.50
- District:Friedrichshain-Kreuzberg

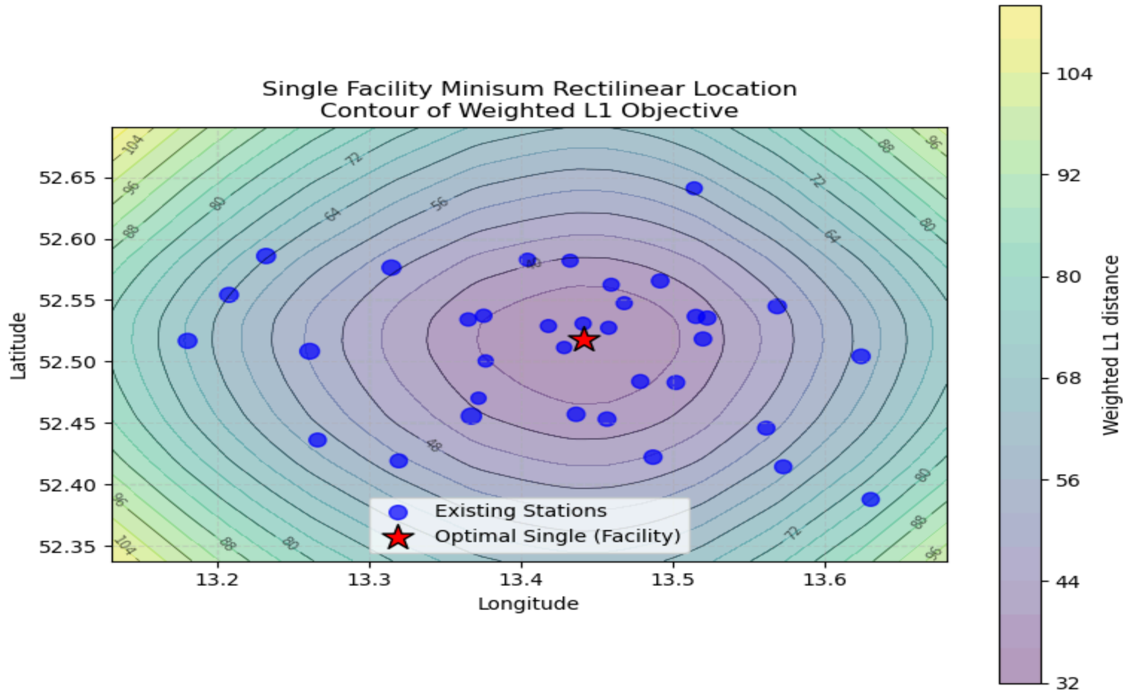

Figure 8. Initial Solution

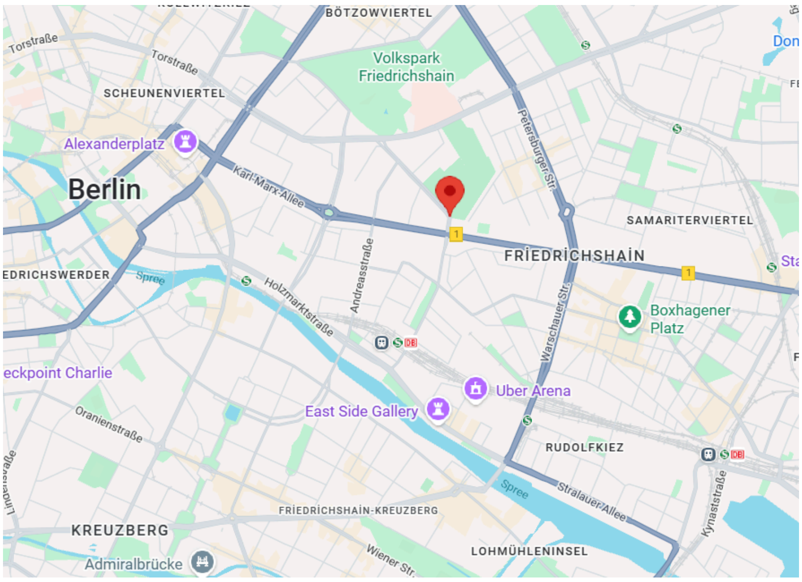

Figure 9. Initial Solution Shown on Real Map

The optimal point obtained from the initial single facility rectilinear distance model lies within the district of Friedrichsain-Kreuzberg. A revenue-cost evaluation was carried out for this candidate location, and under the known demand and price conditions, the point appears to be potentially profitable compared to other districts of Berlin. However, after considering legal and environmental restrictions, it was determined that the optimal points falls inside a public park; therefore, constructing a fuel station at this location is not legally allowed. To avoid such cases, an iterative approach has been implemented, and the model was solved again. A circular forbidden zone with a radius of 11 km was defined around the initial optimal point, this restriction was incorporated into the mathematical model, and the model was repeatedly solved until a legally feasible location was identified.

3.2. First Iteration Solution

In order to visualize the outcome of the advanced model for the first iteration, the rectilinear location model was solved using the existing facility data and initial solution.The map below presents the spatial distribution of the current facilities together with the optimal facility location estimated by the model.

According to the first iteration solution, the model identified the following coordinates as the optimal location for the new facility:

- Latitude:52.516812
- Longitude:13.341861
- Objective Function:39.50
- District:Tempelhof-Schöneberg

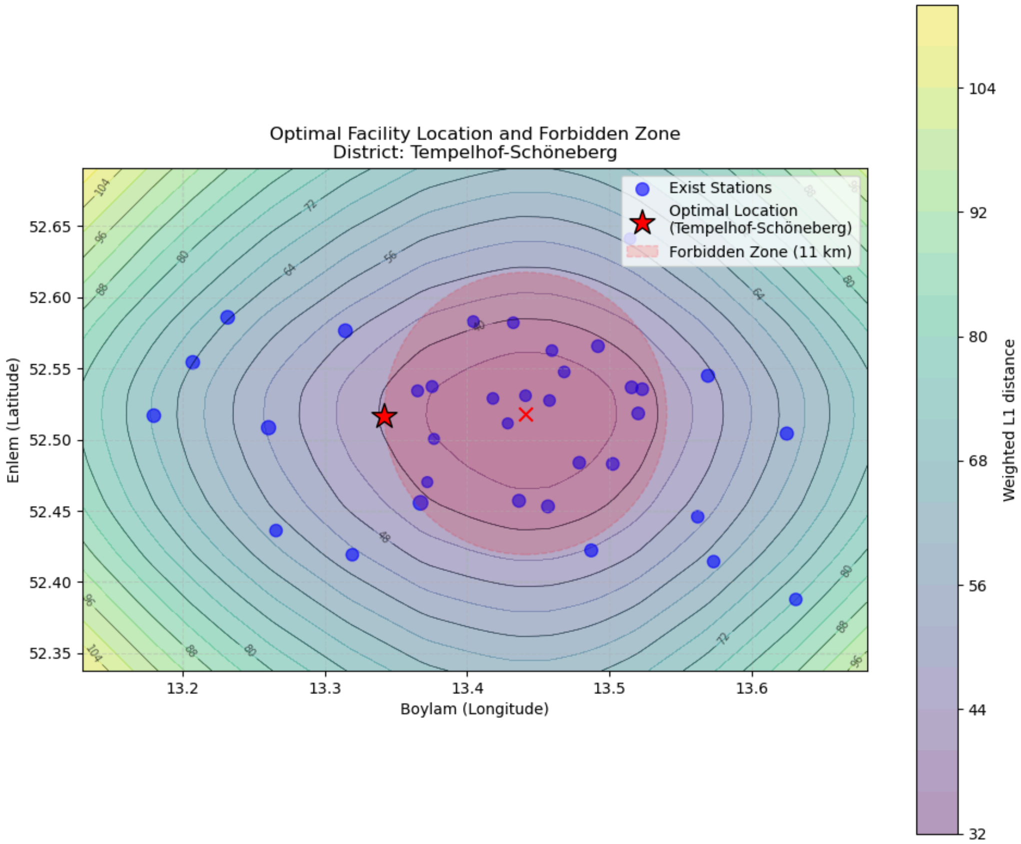

Figure 10. New Solution After First Iteration

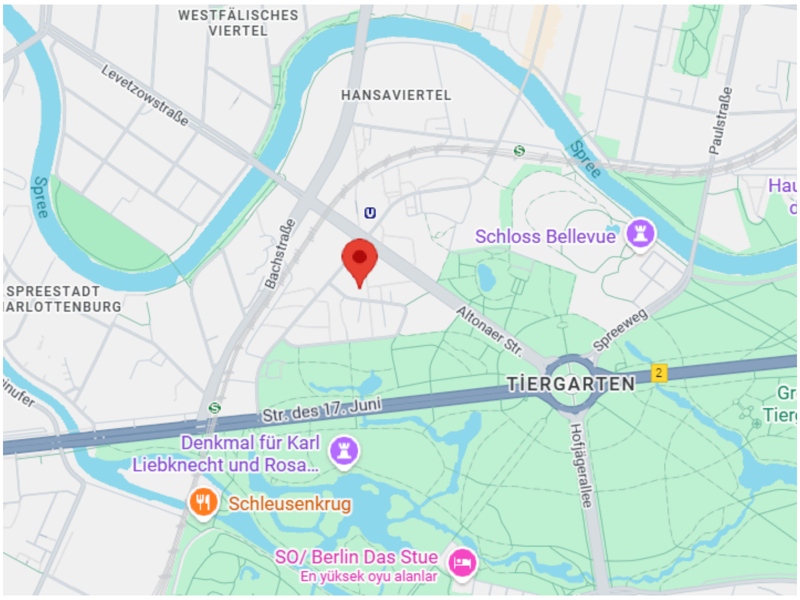

Figure 11. New Solution Shown on Real Map

After re-solving the model with the forbidden zone constraint incorporated, the solution obtained in the first iteration indicates that the new optimal point is located within the district of Tempelhof-Schöneberg. Based on the zoning regulations, environmental requirements, and municipal criteria for fuel-station placement, this area is classified as legally suitable for the construction of a gas station. Furthermore, considering the district's accessibility to major transportation networks, its traffic density and, and the surrounding population distribution, the newly identified location is assessed to be operationally advantageous. Therefore, the solution emerging from the first iteration is regarded as a feasible and implementable alternative both in terms of regulatory compliance and potential commercial profitability.

4. Conclusion and Recommendations for Facility Location

4.1. Conclusion for Facility Location

This study successfully applied a weighted rectilinear distance minimization model to determine the optimal placement of a new fuel station in Berlin. The initial optimal solution, found within the Friedrichshain-Kreuzberg district, was rendered legally infeasible upon environmental and planning assessment, as it fell within a public park.

To align mathematical optimality with real-world constraints, an iterative optimization approach utilizing a circular forbidden-zone constraint was implemented. The model was successfully resolved after incorporating an 11 km exclusion radius around the infeasible region.

The primary finding is that the first iteration of this restricted model yielded a new optimal location within the Tempelhof-Schöneberg district. This alternative site not only achieves compliance with all applicable zoning regulations and environmental constraints but also favorably aligns with critical operational metrics, including traffic density, industrial activity, population distribution, and overall accessibility.

In conclusion, the Tempelhof-Schöneberg location is deemed the definitive and recommended site for development, representing the optimal trade-off between minimizing weighted travel distance and adhering to strict legal and commercial feasibility requirements. This outcome validates the effectiveness of the forbidden-zone methodology in generating reliable, actionable solutions for strategic facility location problems.

4.2. Recommendation for Facility Location

Additional recommendations are suggested in this section.

Contour-Line-Based Alternative Exploration

The generated contour plots provide a rich geographic visualization of the weighted rectilinear distance surface. By systematically analyzing areas with near-optimal objective values, additional candidate regions can be identified and found. These locations may provide practical advantages such as lower construction cost or lower rental cost, easier permitting, or strategic proximity to transportation corridors and highways.

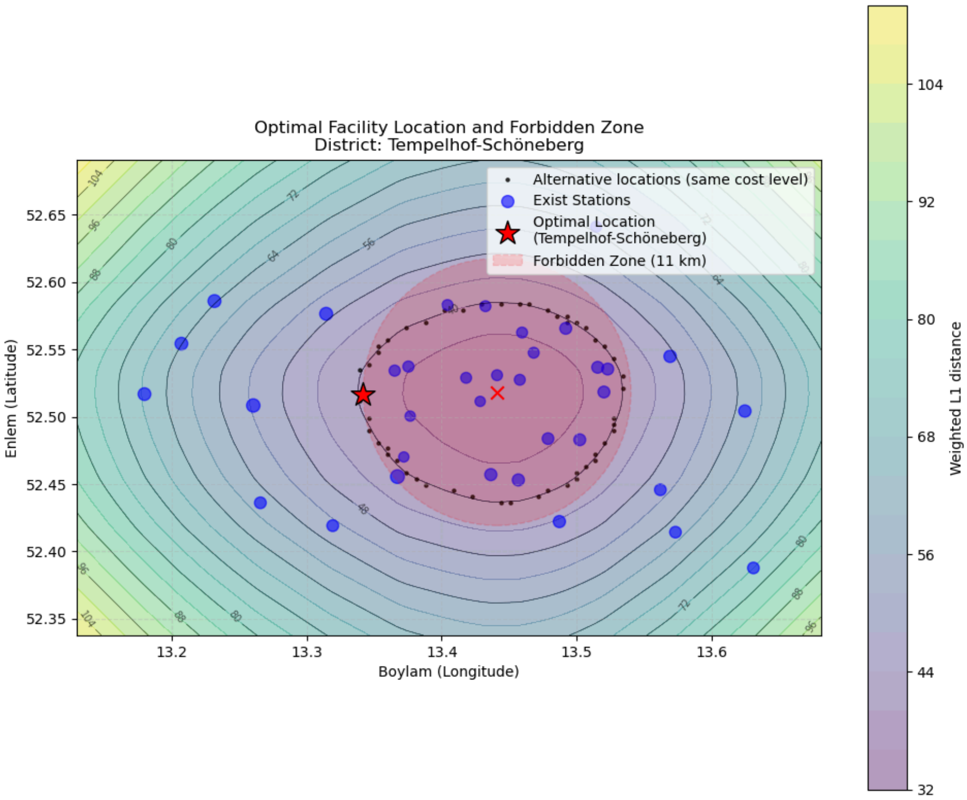

Figure 12. Contours and Optimal Location Under Forbidden Zone Constraints

In Figure 12, the blue points indicate the existing stations, while the colored contour bands visualize the level sets of the weighted rectilinear distance function. Darker contours indicate regions with lower objective values, meaning they result in smaller total weighted distances and are therefore more attractive from the model's perspective. The red star marks the final optimal solution obtained after implementing the forbidden-zone constraint to the model. The additional black points correspond to alternative locations that lie on— or extremely close to— the same contour level. These points show that there exists a set of geometrically different positions that provide almost identical objective values, indicating that the model allows some flexibility in selecting the final location without materially affecting performance.

The shaded circular region represents the legally restricted zone. The visualization shows that some of the near-optimal candidate points fall inside this prohibited area. While these points are mathematically appealing, they are not feasible in practice because they violate the restricted zone constraints. By enforcing the forbidden-zone condition, the model ensures that The final selected point stays outside the restricted area while remaining close to the lowest contour levels. This shows how the zone restriction changes the feasible region and how the model maintains a balance between legal requirements and distance-based optimality, leading to a solution that is both realistic and reliable.

Development of Multiple Facility Optimization Model

Extending the current single-facility model into a multiple-facility (p-median) or multi-facility minisum model would enable evaluation of several potential fuel-station locations. This would allow for:

- evaluating interactions between facilities,
- reducing overall weighted travel distance for the entire network of stations,
- supporting district-level or city-wide infrastructure planning.

Such an extension would offer a more detailed and consistent decision-support system for long-term strategic placement.

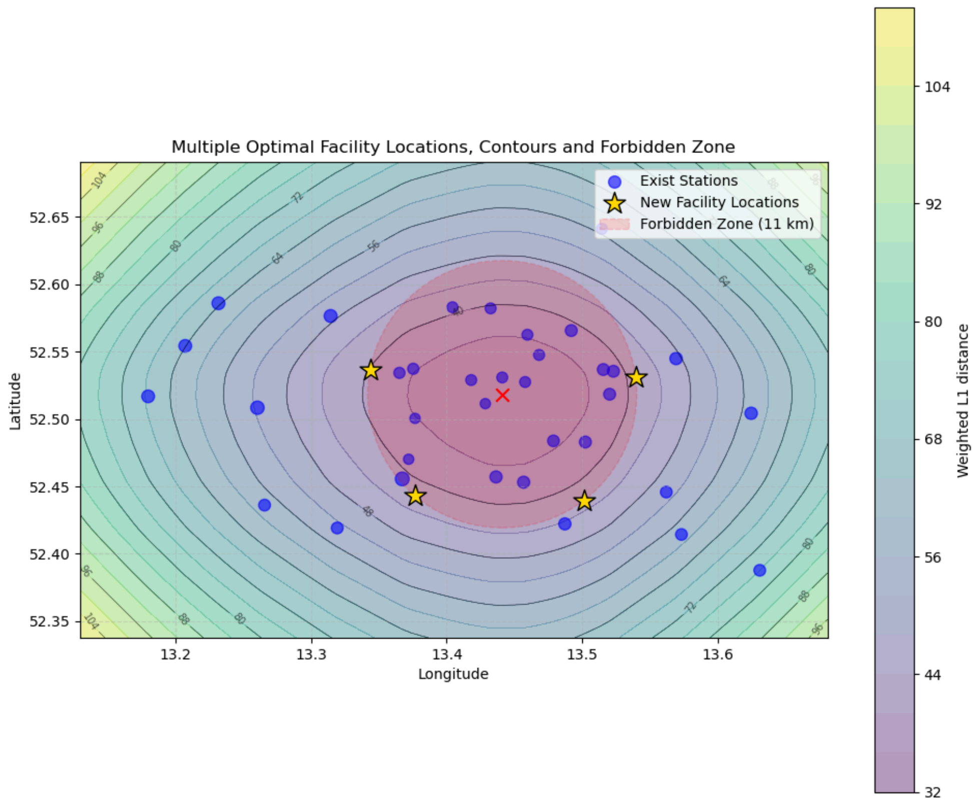

Figure 13. Alternative New Facility Locations

In order to improve the first-iteration analysis, the model was reformulated as a multiple-facility minisum rectilinear distance problem, which made it possible to optimize several new fuel-station locations at the same time. In this extended model, the pairwise distances between the new facilities were intentionally given a weight of zero, using this as an additional analytical approach rather than a full representation of the real system. As a result, the objective function did not penalize the new facilities for being close to each other, and the optimization focused only on minimizing the weighted rectilinear distances between the existing stations and the new candidate locations. This approach separates the effects of the demand distribution and the AHP-based weights, allowing us to clearly observe how the model naturally groups or spreads the new facilities when no interaction costs are included. Although this provides useful insight into the demand structure, it also shows that any desired spacing or operational dependence between the facilities must be added explicitly through extra constraints or penalty terms if needed in later planning stages.
The multiple facility model identified the following coordinates as the alternative locations for the new facilities, achieving an Objective Function value of 165.53.

Table 2. Alternative Solutions

Alternative	Latitude	Longitude	District
1	52.530831	13.539254	Lichtenberg
2	52.443196	13.376386	Tempelhof-Schöneberg
3	52.536691	13.343555	Mitte
4	52.439479	13.500897	Neukölln

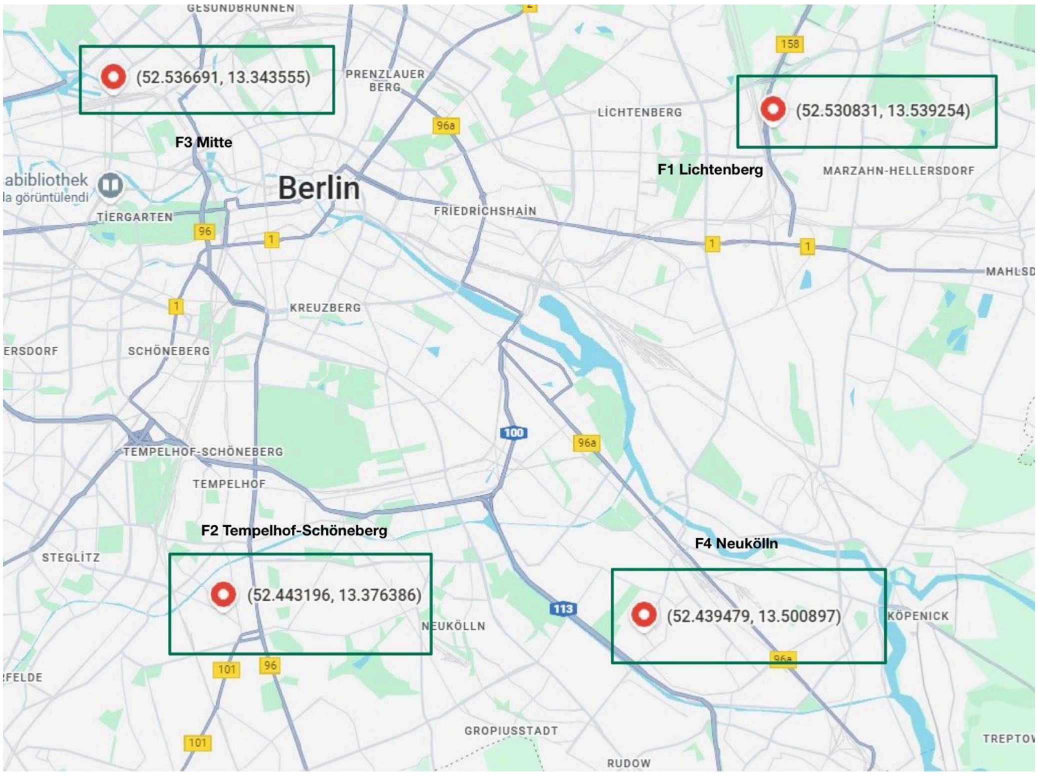

Figure 14. Alternative Facility Locations Shown on Real Map

As a result, instead of iterating multiple times with different forbidden zones, the solution of the multi-facility model can be seen as a set of potential optimal locations. In this study, the model produced four solutions: Facility 1 in Lichtenberg, Facility 2 in Tempelhof-Schöneberg, Facility 3 in Mitte, and Facility 4 in Neukölln. These districts represent areas with high or balanced demand based on the population, traffic, and industrial activity scores from the dataset. The placement of Facility 3 in the Mitte district is especially meaningful because Mitte is located at the center of Berlin, where accessibility, mobility, and service potential are highest. Facility 4 in Neukölln and Facility 1 in Lichtenberg also fit well with significant residential and mixed-use zones, while Facility 2 in Tempelhof-Schöneberg occupies a strategic location close to major transport corridors.

Therefore, the multiple-facility model illustrates a set of strong candidate locations. If only one new station is to be opened, Facility 3 (Mitte) appears to be the most strategically advantageous choice due to its central location and high surrounding demand levels.
If a wider expansion is planned, all four locations can be constructed together.

Scenario-Based Regulatory and Economic Evaluation

The optimization results can be enhanced by evaluating multiple scenarios such as:

- different population-growth projections,
- future traffic-flow forecasts,
- regulatory shifts (e.g., new emission zones),
- varying fuel demand or electric-charging demand.
  This will enable robust decision-making under uncertainty.

GIS(Geographical Integration System)-Integrated Feasibility Screening

Integrating the mathematical model with geographical datasets (land-use maps, green-area limits, protected zones, pipeline and underground infrastructure) will automatically eliminate some of the infeasible regions quickly and reduce the need for iterative forbidden-zone adjustments.Moreover, such an extension is expected to reduce the algorithm’s runtime and enhance overall computational efficiency.

Stakeholder-Oriented Multi-Criteria Evaluation

After identifying feasible optimal points, a secondary multi-criteria assessment including cost, risk, accessibility, environmental impact, and social acceptance can be implemented to refine and enhance the output of the model.

5. Cost Analysis and Financial Feasibility

To ensure the feasibility of the founded facility locations, a detailed cost analysis was constructed. This analysis categorizes costs into two components: Variable Costs, representing the efficiency of the network from a customer perspective, and Direct Financial Costs, covering the initial setup and ongoing operations. This approach helps us see the balance between reaching more customers and spending less money.

5.1 Variable Cost:Transportation and Accessibility Efficiency

In the mathematical facility location model , the main objective is to minimize the weighted travel distance. Although this is not a direct financial cost for the investor, it reflects the customer's "Cost of Inconvenience."

Market Impact: Reducing this cost is closely related to potential revenue. When customers travel shorter distances, they visit more often and the station can attract a wider customer base.

Network Efficiency: This measure also indicates how competitive a station's location is. A site with a low weighted travel distance can serve the highest demand with minimal customer effort.

5.2. Fixed Costs: Initial Investment (CAPEX)

Following the location optimization, a feasibility study was conducted to estimate the Capital Expenditure (CAPEX) required to launch a new station. The setup costs are heavily related to district's real estate characteristics and include:

- Land Acquisition Cost: This is the most variable component and is calculated using the official district-level land values (Bodenrichtwerte in €/m²) for a standard 1,500 m² commercial plot. Central districts like Mitte have much higher prices because land is limited and commercial activity is dense.

- Construction & Infrastructure: A fixed estimation covering the physical station, including:
    - Underground fuel storage tanks and environmental containment systems.
    - Dispensing pumps and canopy structures.
    - Main building construction (shop, cashier area, and sanitary facilities).

- Licensing, Permits & Compliance: Estimated costs for obtaining necessary environmental clearances, construction permits, and municipal safety approvals required for handling hazardous materials.

5.3. Operational Expenses (OPEX)

Beyond the initial setup, the analysis considers the periodic Operational Expenditure (OPEX). These costs vary by region due to differences in local wage expectations, insurance premiums, and business tax rates. The estimated operational expenses include:

- Personnel Costs: Staffing for 24/7 or extended operation hours.
- Utilities & Maintenance: Electricity, water, and regular maintenance of technical equipment (pumps, car wash systems).
- Inventory & Logistics: Holding costs for fuel and retail store inventory.

5.4. Regional Cost Breakdown and Strategic Trade-offs

The financial requirements differ significantly across Berlin's districts. As presented in Table 3, there is a clear cost difference from the city center to the outer districts.

Table 3. Cost Components

Region	Setup Cost(Capex)	Operation Expenses(Opex
Mitte	3.5M € - 4.5M €	75.000 €
Charlottenburg	3.2M € - 4.0M €	70.000 €
F-Kreuzberg	2.8M € - 3.5M €	65.000 €
Neukölln	2.0M € - 2.5M €	50.000 €
Steglitz-Zehl.	2.5M € - 3.0M €	55.000 €
Pankow	1.9M € - 2.4M €	48.000 €
Tempelhof-Sch.	1.8M € - 2.2M €	45.000 €
Treptow-Köp.	1.6M € - 2.0M €	42.000 €
Reinickendorf	1.6M € - 2.0M €	40.000 €
Spandau	1.5M € - 1.9M €	40.000 €
Lichtenberg	1.5M € - 1.9M €	38.000 €
Marzahn-Hell.	1.2M € - 1.6M €	35.000 €

5.5. Strategic Conclusion

The analysis highlights a critical strategic trade-off:

- High-Cost / High-Potential Strategy (Mitte, Charlottenburg): Locations like Facility 3 (Mitte) require the highest initial investment (up to 4.5M €) and operational costs. However, they offer superior brand visibility and access to the highest traffic volume, justifying the premium through higher expected turnover.

- Cost-Efficiency Strategy (Lichtenberg, Marzahn): Locations like Facility 1 (Lichtenberg) present a lower barrier to entry (approx. 1.5M €). While the immediate catchment population may have lower density than the center, the significantly lower CAPEX and OPEX offer a faster Return on Investment (ROI) and reduced financial risk.

Therefore, the optimal network expansion should balance these high-profile central locations with cost-effective peripheral stations to maintain overall portfolio health.

6. Facility Layout Design

This phase of the project focuses on the micro-level design, specifically the internal relations of the newly located TotalEnergies service station. The design must efficiently support four primary functions: traditional fueling, electric vehicle (EV) charging, and convenience store, and additional services. The main objective is to develop a layout that maximizes operational efficiency, customer safety, and provides fluid traffic flow within the site.

6.1. Facility Components

The strategic planning for the new TotalEnergies service station requires defining the operational areas (departments) needed for the design. These components must meet the demands of the market, including fuel, EV charging, and maximizing additional income. These required components form the foundation for the layout analysis that follows.

Core Service Areas

- *Fuel Islands (Traditional Fueling):* This area is for quickly dispensing gasoline and diesel. It must include multiple pumps and a canopy. The design needs to ensure easy access, fast service, and a clear exit to prevent traffic backups.

- *EV Charging Zone (Electric Vehicle):* A separate area for high-speed DC charging units. Because EV charging takes longer than fueling, this zone is a distinct activity center. The design must place charging bays close to customer facilities (Café, Store, Restrooms).

- *Convenience Store (STORE):* This is the main building. It holds the cash register/checkout area, shelves for retail goods, and storage rooms. Its layout is key to maximizing profit from all customers.

Customer Comfort and Support

- *Cafe and Seating Area (CAFE):* This area is either inside or next to the main building. It offers prepared food, drinks, and seating. It is crucial for earning revenue during the longer EV charging wait times and improving the overall customer experience and satisfaction.

- *Restrooms (WC)***:** Customer restrooms that are easy to find and access. They must be located inside or immediately next to the Store building for easy and fast use.

Extra Vehicle Services

- *Vehicle Wash (WASH):* An area for an automatic or self-service car wash. Its location must avoid traffic conflicts with the main fueling and EV zones while supporting a logical flow from other service areas.

- *Air/Water Station (AIR):* A simple, self-service area for filling tires and cleaning windows. It should be positioned for quick use without blocking the main path.

Operations and Management

- *Office Area (OFFICE):* Dedicated space for the site manager, administrative tasks, inventory control, and staff breaks. It should be close to the Store/Cashier area for easy supervision and access to controls.

Traffic Flow and Access

- *Entry (ENTRY):* The point where vehicles enter the facility from the main road. The entry design must allow cars to slow down safely and guide them clearly toward the service zones.

- *Exit (EXIT):* The point where vehicles leave the facility to return to the road. The exit must be separate from the Entry point to keep traffic moving in one direction and ensure safety.

7. Construction Algorithms and Layout Planning Methods

The facility layout phase aims to design the internal spatial configuration of the new TotalEnergies service station in a way that ensures safe traffic flow, operational efficiency, and an enhanced customer experience. To achieve this goal, a structured methodology was implemented, combining REL Chart analysis, ALDEP-based block layout generation, and adjacency–distance evaluation. The service station is planned on a 1200 m² site, and all functional departments were assigned area requirements consistent with realistic service-station dimensions.

Table 4. Departmental Area Requirements

Department	Required Area(m²)
FUEL	315
EV	105
STORE	135
CAFE	120
WC	30
WASH	135
AIR	30
OFFICE	45
ENTRY	143
EXIT	142
TOTAL	1200

Since the total available site area is 1200 m², this full area is used as the effective operational space in the ALDEP layout model.

Accordingly, the ALDEP algorithm is applied to a 30 × 40 grid (1200 cells), where each cell corresponds to 1 m².

7.1. Relationship (REL) Chart Construction

A REL Chart was developed to express qualitative closeness requirements between departments such as Fuel, EV Charging, Store, Café, WC, Wash, Office, Air/Water, Entry, and Exit.

Standard closeness codes (A, E, I, O, U, X) were used to reflect operational dependencies such as:

- A (Absolutely Necessary)
- E (Especially Important)
- I (Important)
- O (Ordinary)
- U (Unimportant)
- X (Undesirable)

All relationships were transformed into numerical scores using standard ALDEP weights, enabling quantitative comparison between layout alternatives. The numerical scores have been given on Table 5.

Table 5. Numerical Scores

RELATIONSHIP	**VALUE**
A	64
E	16
I	4
O	1
U	0
X	-1024

Based on the operational requirements, customer behavior patterns, safety considerations, and the functional analysis previously discussed, a complete Relationship Matrix and REL Chart were constructed for the facility layout. The matrix reflects the qualitative closeness needs among all departments using the standard SLP (Systematic Layout Planning) codes (A, E, I, O, U, X).

From the full matrix, the ten most critical relationships that significantly influence layout decisions are summarized below.

FUEL – ENTRY (A: Absolutely Necessary): The fueling area must be immediately accessible once vehicles enter the station. This reduces confusion, improves traffic flow, and prevents congestion at the entrance.

EV – CAFE (A: Absolutely Necessary): EV charging requires longer waiting times. Therefore, locating the Café near the EV zone significantly improves customer satisfaction and increases revenue opportunities during charging periods.

STORE – WC (A: Absolutely Necessary): Customers expect restrooms to be easily accessible from inside the store. This adjacency is essential for customer convenience and aligns with standard service-station design principles.

WASH – EXIT (A: Absolutely Necessary): Vehicles typically proceed to the exit after using the car wash. Placing the wash area close to the exit ensures a one-directional, conflict-free traffic flow.

FUEL – STORE (E: Especially Important): Fueling customers frequently visit the store for payment, snacks, or services. Locating these two areas relatively close reduces unnecessary walking distance and encourages impulsive purchases.

EV – STORE (E: Especially Important): EV users spend additional time on-site. Placing the store near the charging bays encourages revenue generation during charging waits.

STORE – OFFICE (A/E: Highly Important): The office must supervise store operations, cash-handling processes, and inventory management. Close proximity ensures operational control and staff efficiency.

CAFE – WC (A: Absolutely Necessary): Restrooms should be located near the café seating area to meet hygiene expectations and maintain customer convenience.

FUEL – WASH (E: Especially Important)*:* A common customer sequence is “fuel-car wash.” Placing these areas close supports natural vehicle flow and reduces internal traffic conflicts.

ENTRY – EXIT (X: Undesirable): Entry and exit points must remain spatially separated to avoid dangerous counter-flows. This prevents accidents and ensures smooth directional circulation.

To support the analysis, a relationship heat map (Figure 15) and REL Chart (Figure 16) were also generated , visually indicating the intensity of interactions between departments.

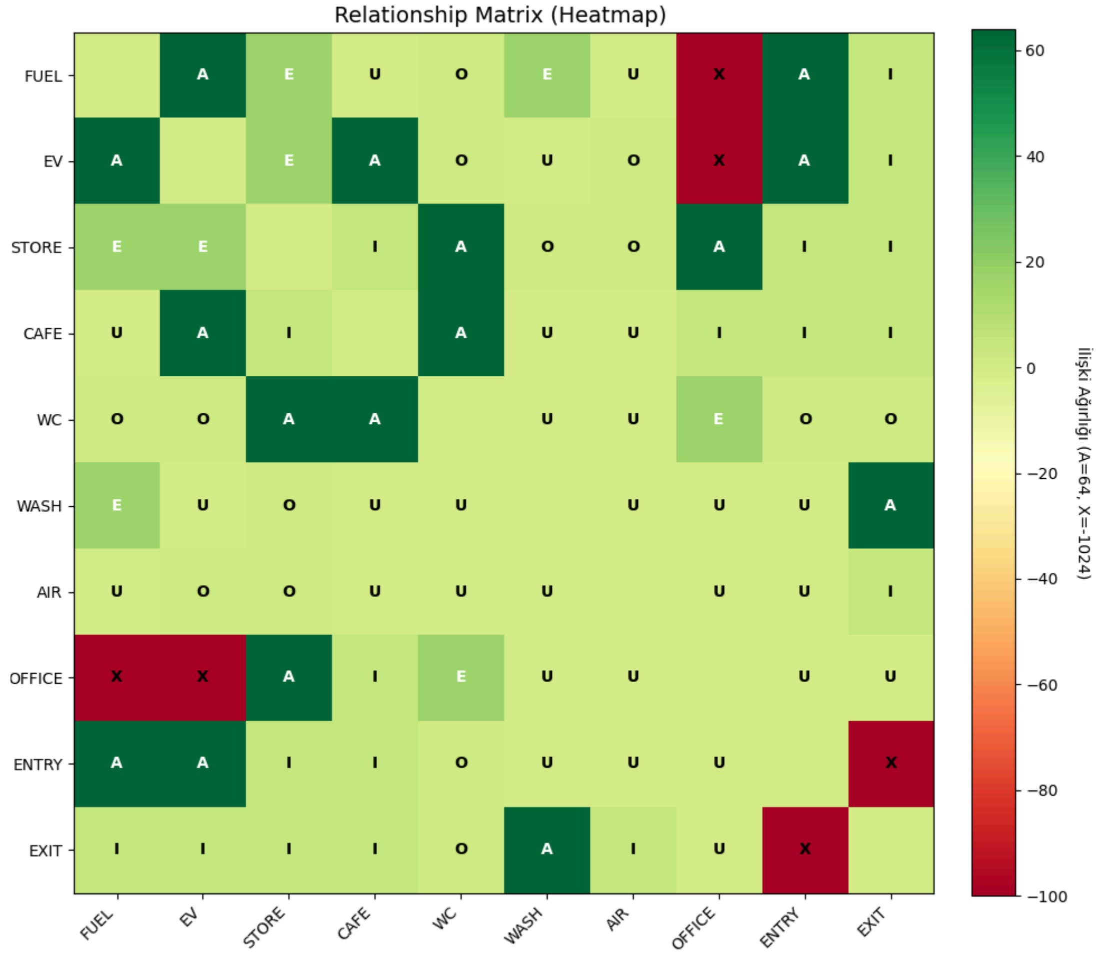

Figure 15. Relationship Matrix (Heatmap)

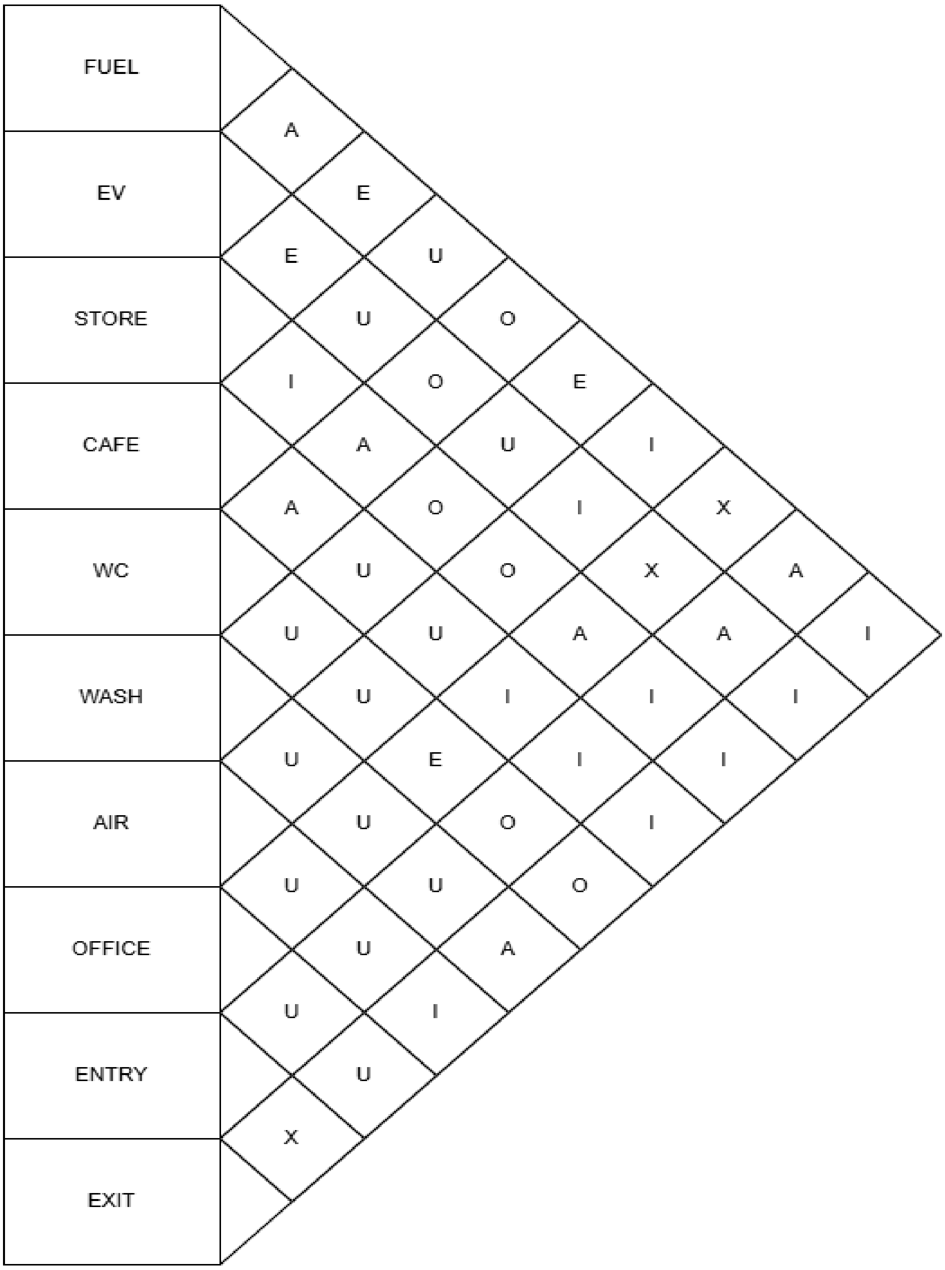

Figure 16. REL Chart

7.2 ALDEP Construction and Layout Generation

Following the development of the Relationship Matrix and REL Chart, the next step in the methodology is the construction of block-layout alternatives using the ALDEP (Automated Layout Design Program) algorithm. ALDEP is a classical heuristic which is widely used in Systematic Layout Planning (SLP) to generate feasible block layouts when requirements and relational closeness scores are decided.

As established in the Methods section, the total area of the fuel station is 1200 m^2, fully allocated to functional departments.

To implement area to the relatable software programs:

- A 30x40 grid was defined
- Each grid cell corresponds to 1 m^2
- Department areas were mapped directly onto this grid
- ALDEP uses these exact areas to place departments proportionally

The ALDEP Algorithm used in this study follows two main principles:

Adjacency-Based Evaluation

After each layout is generated, ALDEP computes an adjacency score based on the numerical values converted from the REL Chart Adjacency-Based model and its components are given below.

$$w_{ij} \in \{64, 16, 4, 1, 0, -1024\}$$

I. $Sets$
$D = \{1,....., m\}: number\ of\ departments$

II. $Parameters$
$w_{ij}\ (i, j) \in (D\ x\ D): Importance\ Score\ between\ department\ and\ department\ j$

III. $Decision\ Variables$
$x_{ij}(i, j) \in (D\ x\ D): \{1\ if\ departments\ p\ and\ q\ are\ adjacent\ ,\ 0\ otherwise\}$

IV. $Objective\ Function$

$$min\ Z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij} x_{ij}$$

Distance-Based Evaluation

In addition to the adjacency-based analysis, a second quantitative evaluation method is used to assess the quality of each layout: the Distance-Based Objective Model. This model measures the total material handling or movement cost implied by the arrangement of departments. Unlike adjacency scoring (which reflects qualitative closeness preferences from the REL Chart), the distance-based model evaluates layouts using actual geometric distances between departments.

The model assumes that each pair of departments has an associated flow value, representing the relative intensity of interaction between them. The goal is to minimize the total weighted travel cost within the facility. Distance-Based model and its components are given below.

I. $Sets$
$D = \{1,....., m\}: number\ of\ departments$

II. $Parameters$

$f_{ij}(i, j) \in (D \times D)$: $flow\ value\ between\ department\ i\ and\ j$

$c_{ij}(i, j) \in (D \times D)$: $Unit\ cost\ of\ movement\ between\ department\ i\ and\ department\ j$

$d_{ij}(i, j) \in (D \times D)$: $Rectilinear\ distance\ between\ the\ centroids\ of\ departments\ i\ and\ j$

III. $Decision\ Variables$

$No\ decision\ variable\ is\ required,\ because\ distance\ evaluation\ is\ performed.$
$Distanced\ d_{ij}\ are\ computed\ directly\ from\ the\ coordinates\ of\ the\ block\ layout.$

IV. $Objective\ Function$

$$min\ Z = \sum_{i=1}^{m}\sum_{j=1}^{m} f_{ij}c_{ij}d_{ij}$$

7.3. Generation of Flow Matrix And Cost Matrix

For the distance-based objective model, two numerical structures are required:

1. a flow matrix, representing the interaction intensity between departments
2. a cost matrix, representing the unit cost of movement

These matrices allow the model to compute the total movement cost across the layout.

Flow Matrix Construction

The flow matrix f_{ij} quantifies how frequently departments i and j interact.

Since no real traffic or material-handling dataset exists for a fuel station, a synthetic flow matrix was generated using the qualitative insights from the REL Chart.

The logic is:

- Stronger REL relationships → Higher simulated flow
- Weaker REL relationships → Lower flow
- Undesirable (X) pairs → Zero or near-zero flow

This preserves the behavioral patterns implied by the REL Chart while enabling quantitative analysis.

Table 6. Flow Matrix

	Fuel	Ev	Store	Cafe	Wc	Wash	Air	Office	Entry	Exit
Fuel	0	0	450	0	0	50	50	0	0	50
Ev	0	0	150	50	0	0	20	0	0	30
Store	0	0	0	170	170	0	0	0	0	450
Cafe	0	0	50	0	50	0	0	0	0	100
Wc	0	0	100	50	0	0	0	0	0	50
Wash	0	0	25	0	0	0	15	0	0	110
Air	0	0	0	0	0	0	0	0	0	150
Office	0	0	10	0	0	0	0	0	0	0
Entry	600	250	75	0	0	40	35	0	0	0
Exit	0	0	0	0	0	0	0	0	0	0

Cost Matrix Construction

The cost matrix c_{ij} specifies the cost per unit distance between departments.
For this project, a uniform cost structure was assumed:

- $c_{ij} = 1$ for all department pairs
- $c_{ij} = 0$ for the diagonal (a department has no cost with itself)

This simplification is commonly used when:

- No special material movement cost differences exist,
- The objective is to compare layouts rather than compute real economic costs.

The cost matrix ensures that differences in the objective value come only from distances and flow, not arbitrary cost weights.

Table 7. Cost Matrix

	Fuel	Ev	Store	Cafe	Wc	Wash	Air	Office	Entry	Exit
Fuel	0	1	1	1	1	1	1	1	1	1
Ev	1	0	1	1	1	1	1	1	1	1
Store	1	1	0	1	1	1	1	1	1	1
Cafe	1	1	1	0	1	1	1	1	1	1
Wc	1	1	1	1	0	1	1	1	1	1
Wash	1	1	1	1	1	0	1	1	1	1
Air	1	1	1	1	1	1	0	1	1	1
Office	1	1	1	1	1	1	1	0	1	1
Entry	1	1	1	1	1	1	1	1	0	1
Exit	1	1	1	1	1	1	1	1	1	0

7.4. Generation of Four ALDEP Layout Alternatives

To avoid bias from a single starting sequence, ALDEP was executed four times, each using a different seed ordering:A sweep width of 3 was selected for all iterations to ensure proportional area distribution across the 30 x 40 grid.

1. A sequence starting with Entry:
   Entry-Fuel-Ev-Cafe-Wc-Store-Office-Air-Exit-Wash
2. A sequence starting with Fuel:
   Fuel-Entry-Ev-Store-Cafe-Wc-Office-Air-Wash-Exit

3. A sequence starting with Store:
   Store-Cafe-Wc-Fuel-Ev-Entry-Air-Wash-Exit-Office
4. A sequence starting with Ev:
   Ev-Fuel-Entry-Store-Cafe-Wc-Office-Air-Exit-Wash

The spatial configurations of the four ALDEP layout alternatives, generated using the different seed sequences, are provided in the figures below.

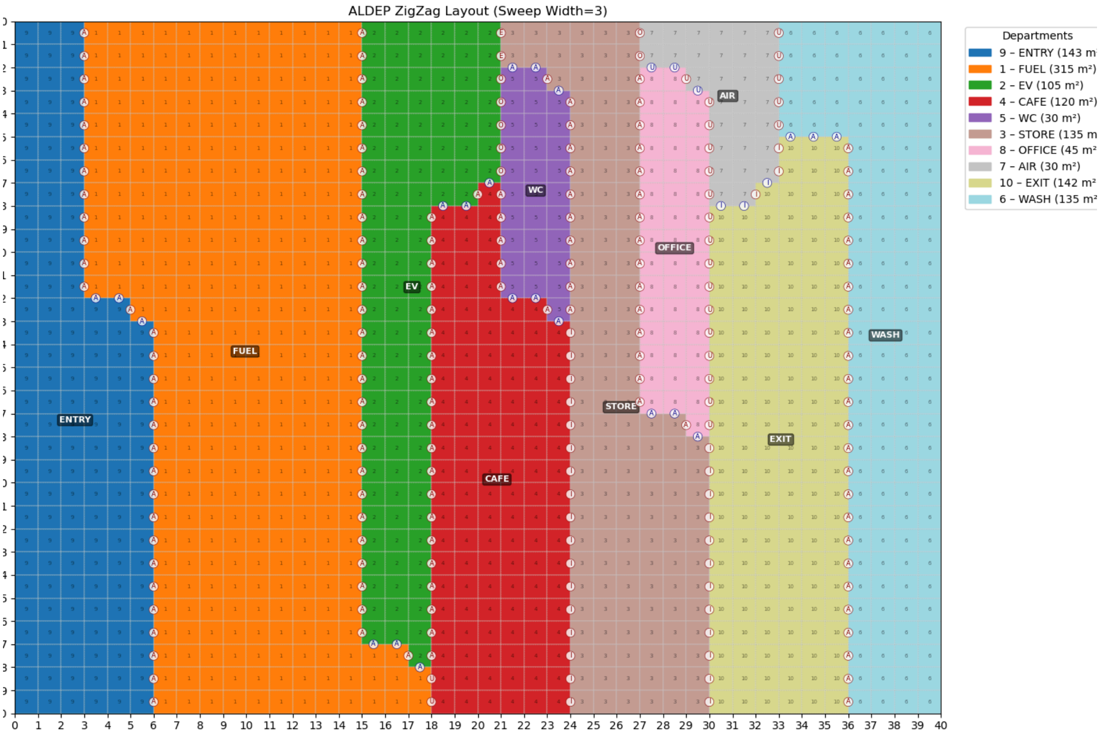

Figure 17. Alternative Layout 1

Adjacency Score of Alternative 1: 956 – Distance Score of Alternative 1: 50025.92

This layout achieves the highest adjacency score among all alternatives, indicating that almost all critical A and E relationships (such as Fuel–Entry and EV–Cafe) are successfully satisfied. However, this strong relational performance is obtained at the expense of a slightly higher total distance cost. It provides the best overall operational layout with the highest safety level and strongest compliance with critical relational requirements.

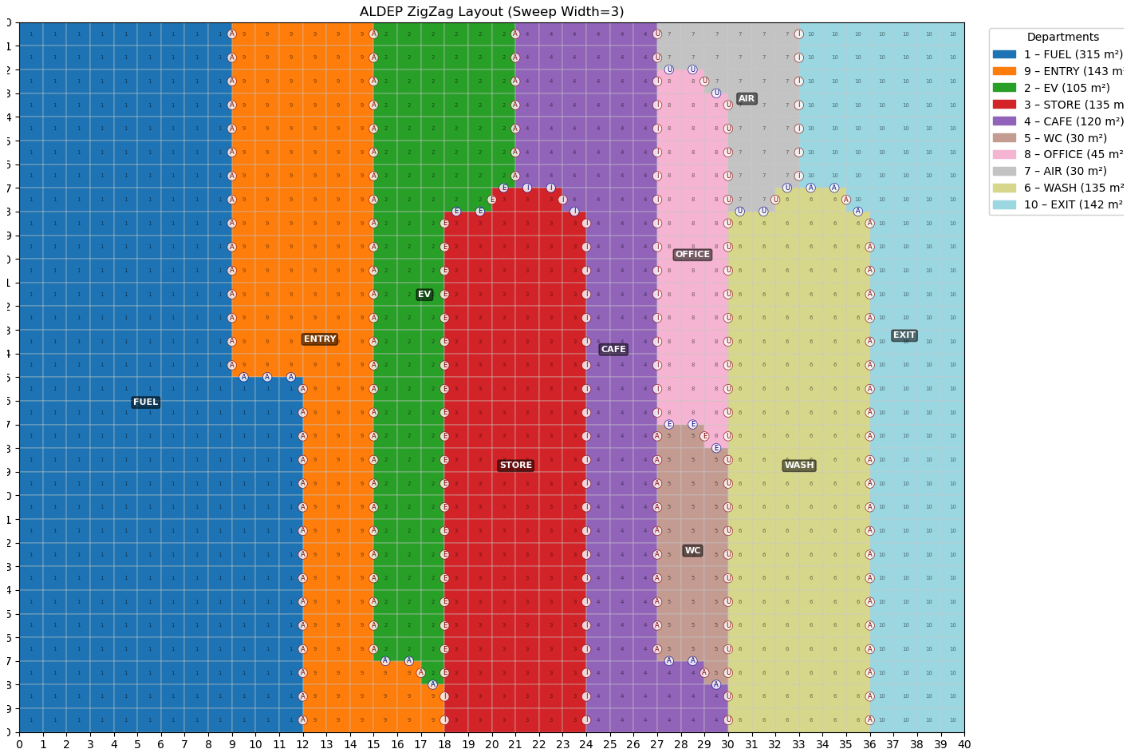

Figure 18. Alternative Layout 2

Adjacency Score of Alternative 2: 736 – Distance Score of Alternative 2: 49900.52

This alternative demonstrates a relatively balanced performance in terms of both adjacency and distance scores. Nevertheless, it fails to satisfy as many high-priority relational requirements as Alternative 1, which limits its overall operational effectiveness. Although it offers a balanced performance in both adjacency and distance criteria, it is not the optimal solution.

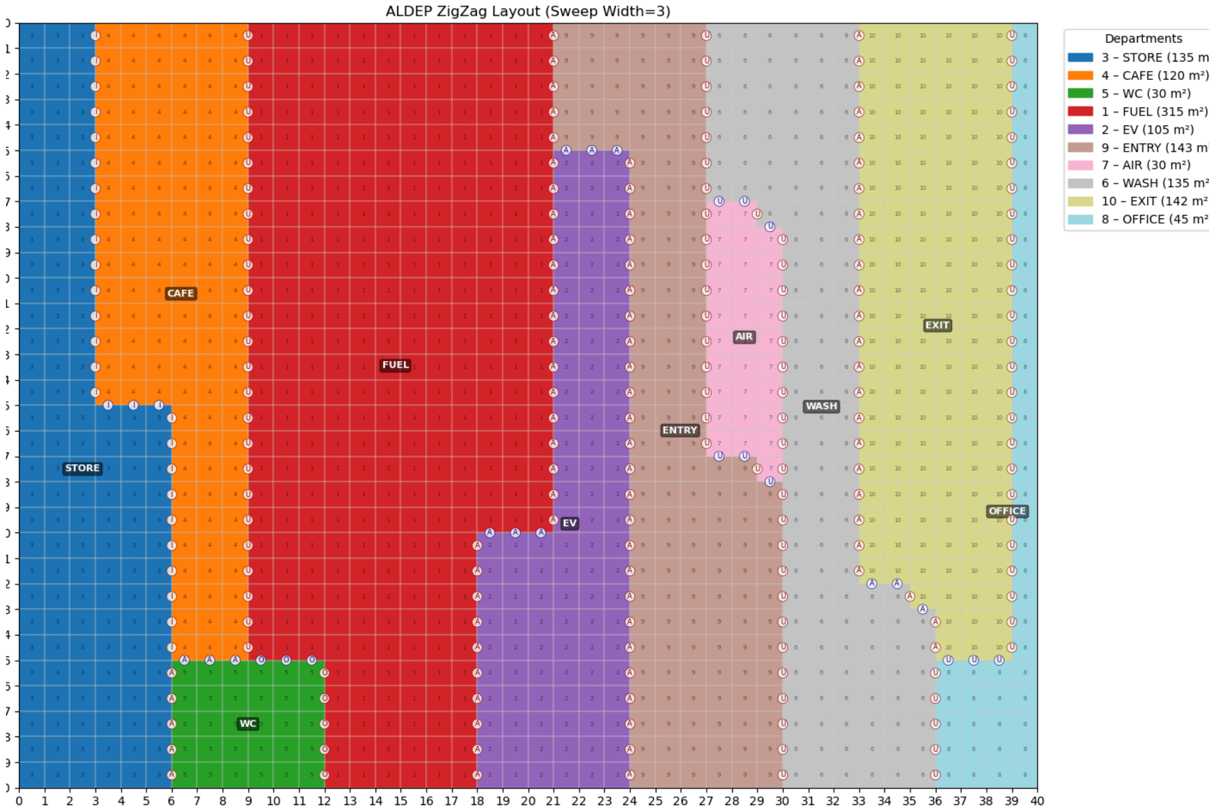

Figure 19. Alternative 3

Adjacency Score of Alternative 3: 778 – Distance Score of Alternative 3: 63055.99

Although this layout attains a moderate adjacency score, it exhibits the worst distance performance among all alternatives. The excessive separation between several important departments results in long internal travel paths and significant operational inefficiency. In summary, it is not suitable due to its weakest performance in both operational efficiency and internal distance structure.

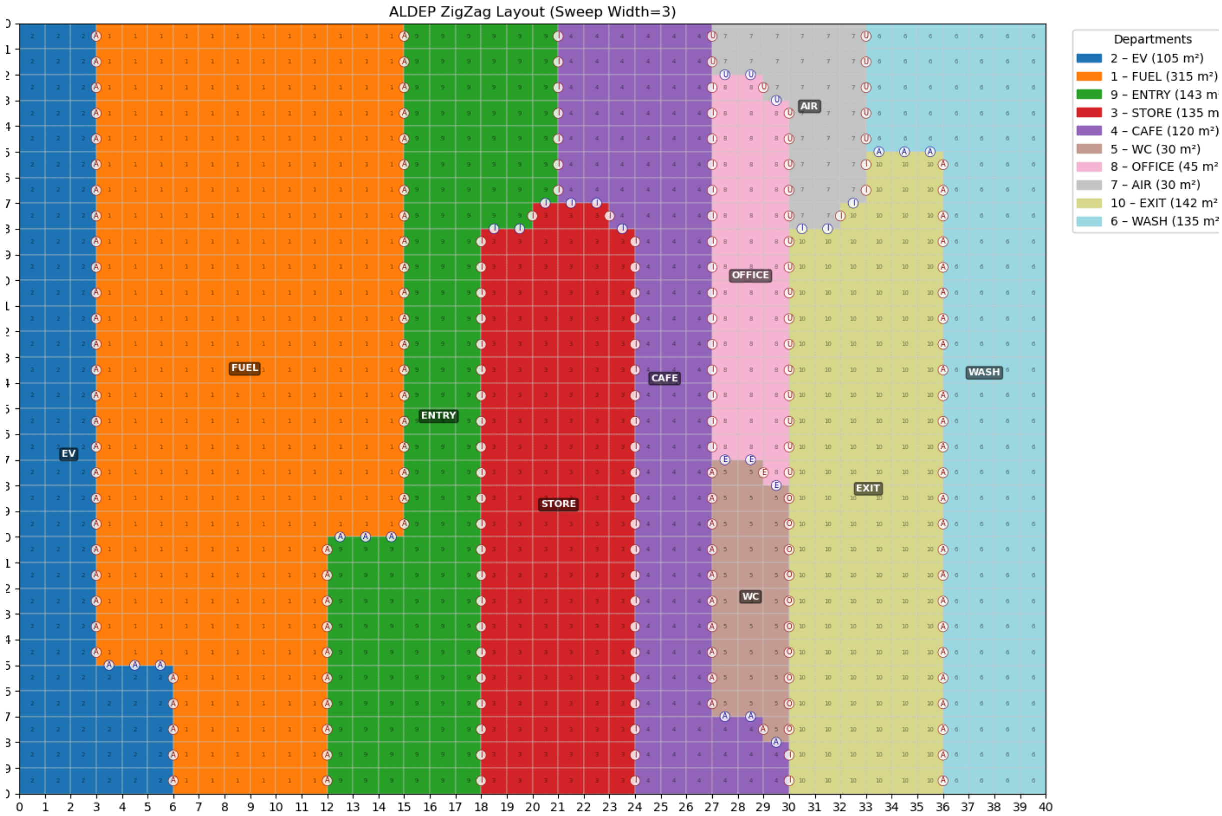

Figure 20. Alternative 4

Adjacency Score of Alternative 4: 594 – Distance Score of Alternative 4: 48839.27

This layout yields the lowest distance-based cost and is therefore the most efficient in terms of total internal movement. However, it significantly compromises several essential adjacency requirements, which weakens the overall operational logic and safety of the facility. It achieves the best distance efficiency but shows weaker performance in terms of essential adjacency relationships.

Although the sweep width, grid size, and department areas remain fixed, varying the initial sequence causes ALDEP to produce distinct block configurations. These alternative layouts allow the designer to explore how different operational priorities influence the geometric arrangement.

8. Results Outputs and Recommendations of ALDEP Layout Alternatives

Four alternative block layouts were constructed using the ALDEP (Automated Layout Design Program) algorithm. These layouts utilized fixed department areas, grid size, and sweep width, but changed the initial department sequencing to prevent bias and explore different geometric arrangements.

These alternatives were evaluated and compared using three criteria:

1. Adjacency Score: Measures the qualitative satisfaction of closeness defined in the REL Chart. A higher score indicates better alignment with functional and safety requirements.
2. Distance-Based Cost: Measures quantitative efficiency by calculating the total weighted rectilinear distance between department centroids. A lower score indicates more efficient customer and vehicle movement for the station.
3. Practical/Operational Feasibility: Evaluates real-world suitability concerning traffic flow logic, safety, accessibility, and construction constraints .

The numerical performance of the four ALDEP layouts and REL Chart relations are summarized below in Table 8 and Figure 21.

Table 8. Numerical Performances of Alternative Layouts

Alternative Number	Adjacency Score	Distance Score
1	956	50025.92
2	736	49900.52
3	778	63055.99
4	594	48839.27

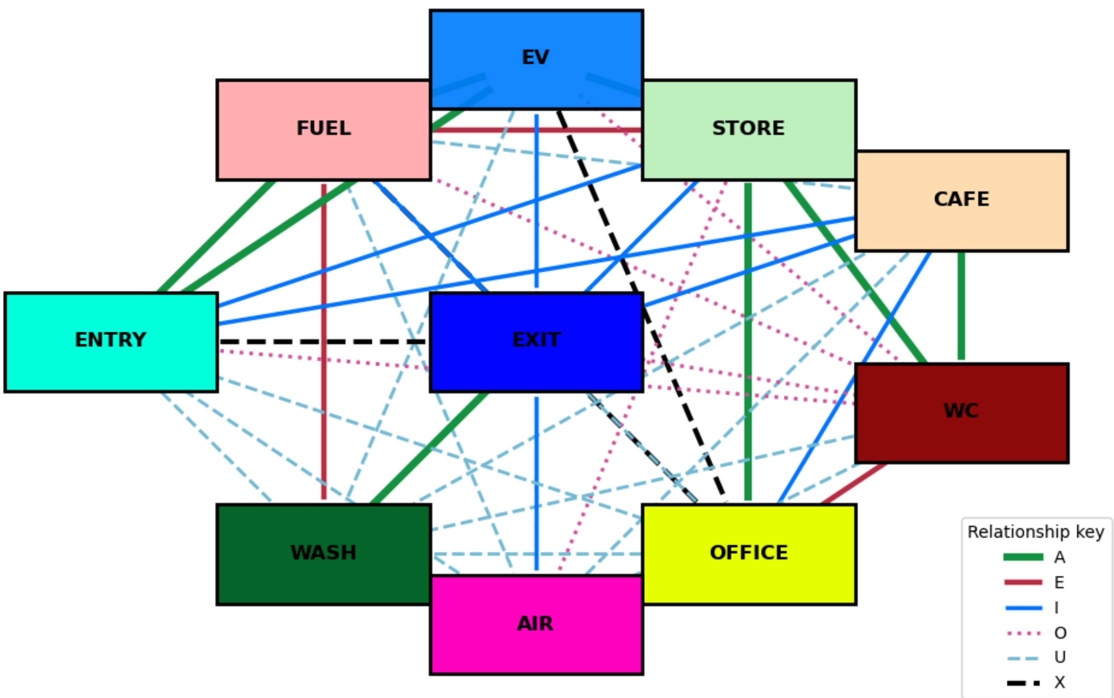

Figure 21. REL Chart Relations

8.1. Comparative Analysis and Selection

The comparison clearly showed a standard trade-off in facility design. Alternative 1 had the highest Adjacency Score (956) , meeting critical priorities like placing Fuel near Entry and EV near the Cafe. This layout is vital for smooth vehicle routing and natural service flow. Conversely, Alternative 4 yielded the lowest Distance Cost (48839.27) , making movement efficient, but it had to sacrifice some essential qualitative closeness . Alternative 2 was the most balanced option, ranking second in both Adjacency (736) and Distance (49900.52). Although usable, it failed to meet as many key priorities as Alternative 1. Alternative 3 was unsuitable, scoring lowest in both metrics. Key operational relationships were separated, leading to longer travel times and highly inefficient operation.

8.2. Selection of Final Layout

For a fuel station, where safety, clear traffic flow, and operational logic are important, the satisfaction of key relational requirements (Adjacency Score) is considered more critical than marginal differences in overall distance cost(Distance Score). Therefore, based on the multicriteria evaluation, the strong match with operational priorities makes Alternative 1 the most appropriate and practical design for real implementation.

Table 9. Summary of Layout Rankings and Final Selection Analysis

Alternative	Adjacency Ranking	Distance Ranking	Overall
Alternative 1	1 (Best)	3	Best operational layout, highest safety and relational compliance
Alternative 2	2	2	Balanced but not optimal
Alternative 3	4 (Worst)	4 (Worst)	Not suitable
Alternative 4	3	1 (Best)	Best distance efficiency, weaker adjacency

8.3. Recommendations for Advanced Layout Optimization

To enhance the depth of the analysis, further computational improvements are necessary:

- Implementation of the CRAFT Algorithm: The block layouts produced by ALDEP are initial feasible solutions. Applying an improvement heuristic like CRAFT (Computerized Relative Allocation of Facilities Technique) is the logical next step to refine the final arrangement by iteratively minimizing the calculated weighted travel cost.

- Varying ALDEP Sweep Width: The current ALDEP iterations were conducted using fixed parameters such as fixed sweep width. Varying the sweep width parameter would allow the algorithm to explore a range of distinct spatial configurations, which is essential for preventing potential bias and ensuring a more comprehensive and enhanced search across the solution space.
- Multi-Objective Integration: Future work should focus on integrating the Adjacency Score and Distance-Based Cost objectives into a single weighted multi-objective model. This would allow for a mathematically optimized design that strategically balances operational logic (qualitative needs) with movement efficiency (quantitative movement cost).

9. Facility Layout Improvements

9.1. Improvement with Sweep Width

To quantitatively support the recommendation for varying the sweep width (SW), a sensitivity analysis was conducted on the final selected layout sequence (Alternative 1, starting with Entry). The model, which was initially executed with a fixed sweep width of 3, was re-run by setting the sweep width to 2 and 4. The results on Table 10 demonstrate how the single parameter change significantly impacts both objective scores.

Table 10.Comparative Performance of Alternative 1 Under Different Sweep Width Scenarios

Alternative	**Sweep Width**	**Adjacency Score**	**Distance Score**
1	2	922	42663.52
1	3(default version)	956	50025.92
1	4	958	49326.82

The results confirm that the solution is highly sensitive to the Sweep Width (SW) parameter. The analysis found that the initial fixed setting (SW=3), which generated the selected Alternative 1 (Score: 956), did not yield the lowest Distance Cost.

By changing the parameter to SW=2, the algorithm produced the best possible efficiency, achieving the lowest Distance Cost of 42663.52. However, this efficiency came at the expense of qualitative requirements, as SW=2 produced the lowest Adjacency Score (922) among the three tests. Conversely, SW=4 yielded the highest Adjacency Score (958) but a higher Distance Cost.

This clear trade-off proves that the initial fixed parameterization missed better results, specifically the most efficient arrangement. This finding strongly reinforces the necessity for comprehensive parameter testing and justifies the recommendation to implement variable sweep width searches to find the most efficient and optimal final layout.

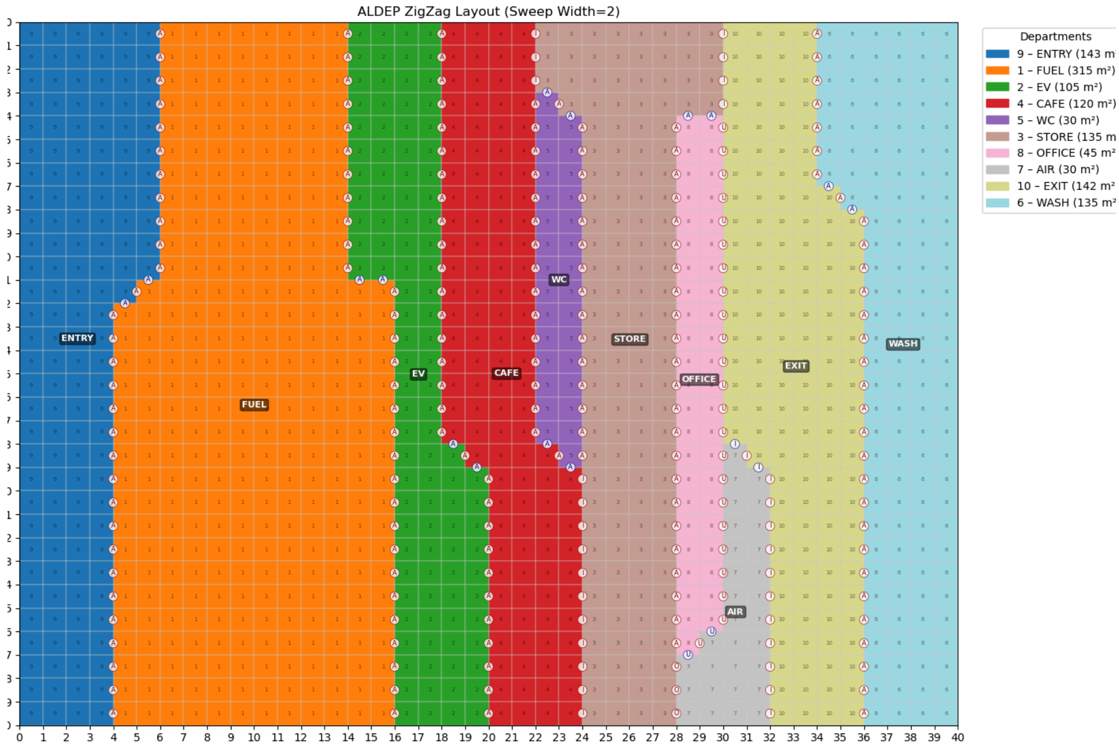

Figure 22. ALDEP layout with 2 unit sweep width

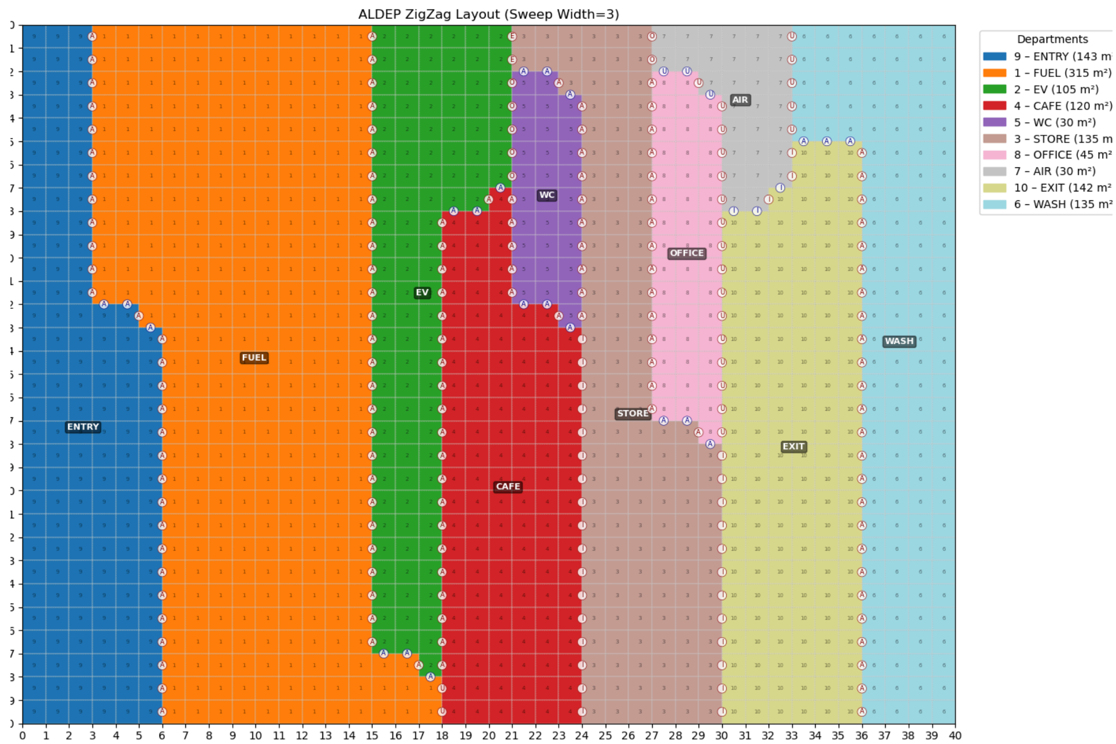

Figure 23. ALDEP layout with 3 unit sweep width

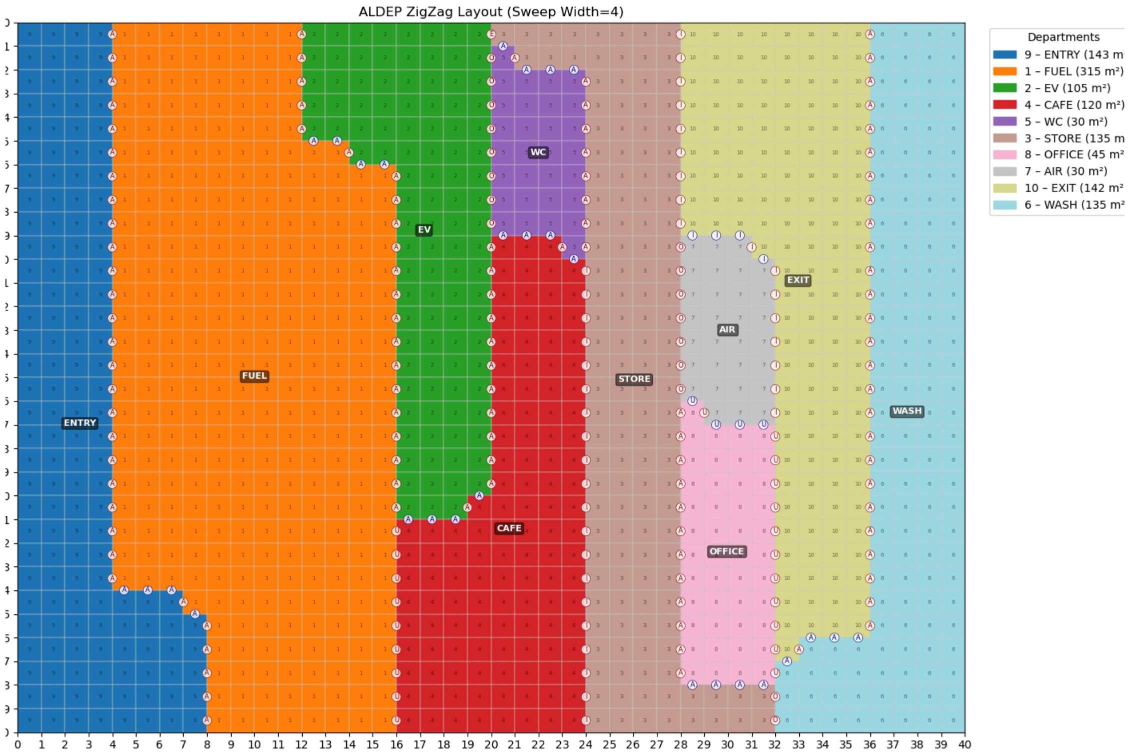

Figure 24. ALDEP layout with 4 unit sweep width

9.2. Layout Optimization with CRAFT

Following the selection of Alternative 1 (generated with Sweep Width=3) as the most practical and relationally compliant design, the next phase focuses on refining its geometric arrangement to maximize efficiency. This refinement utilizes the CRAFT (Computerized Relative Allocation of Facilities Technique) improvement heuristic.

The objective of the CRAFT application is to minimize the total weighted travel cost (Distance Cost) within the facility, using the existing layout as the starting solution.

CRAFT Methodology

- Objective Function: CRAFT operates by iteratively minimizing the distance objective function, which relies on flow between departments cost between departments and rectilinear distance between departments:

$$\min Z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij} c_{ij} d_{ij}$$

$$m \in R : \text{total number of the departments}$$

- Data Inputs: The optimization process will use the synthesized Flow Matrix f_{ij} and the Cost Matrix c_{ij}, which were constructed in Section 7.3 to reflect the operational dependencies and interaction intensity between departments.

- Pairwise Exchange: CRAFT employs the pairwise exchange procedure. This routine systematically evaluates the potential benefit (cost reduction) gained by swapping the locations of two departments that are either adjacent or non-adjacent.

- Iterative Refinement: The algorithm will continue to make swaps, accepting any exchange that leads to a reduction in the total distance objective function. The process stops when no further two-way exchange can reduce the total cost, yielding a locally optimized final layout.

CRAFT Application and Iterations for Alternative 1

Following the methodology defined in CRAFT Methodology Section, the CRAFT algorithm was applied to the selected initial layout, Alternative 1 (Sweep Width=3), which had an initial Distance Cost of 50025.92. The objective was to minimize this cost by iteratively performing the pairwise exchange routine. Table 11 summarizes the convergence process and the total cost reduction achieved:

Table 11. CRAFT methodology results

Iteration	Distance Score	Improvement(%)
Initial	50025.92	N/A
1	46820.516	6.41%
2	43830.3821	6.38%
3	41761.21875	4.95%
4	40009,132	4.20%
5	38899,5	2.77%

Iteration Sequences

The sequence adjustments made by the CRAFT algorithm during the optimization process were as follows:
Initial: Entry-Fuel-Ev-Cafe-Wc-Store-Office-Air-Exit-Wash
Iteration 1: Fuel-Entry-Ev-Cafe-Wc-Store-Office-Air-Exit-Wash
Iteration 2: Fuel-Entry-Ev-Cafe-Wc-Store-Office-Exit-Air-Wash
Iteration 3: Fuel-Entry-Ev-Wc-Cafe-Store-Office-Exit-Air-Wash
Iteration 4: Fuel-Entry-Ev-Wc-Store-Cafe-Office-Exit-Air-Wash
Iteration 5: Fuel-Entry-Ev-Wc-Store-Cafe-Air-Exit-Office-Wash

Optimization Results

The CRAFT optimization process successfully converged in 5 iterations, reducing the total weighted travel cost from the initial value of 50025.92 to a final optimized value of 38899.50. This optimization step resulted in a total cost reduction of 11126.42, representing a significant efficiency gain of approximately 22.24% compared to the original ALDEP-generated layout.The final optimized layout, designated as the CRAFT Final

Solution (Sequence: Fuel-Entry-Ev-Wc-Store-Cafe-Air-Exit-Office-Wash), provides the definitive, most efficient geometric configuration, ensuring maximum compliance with both qualitative requirements and quantitative movement efficiency.

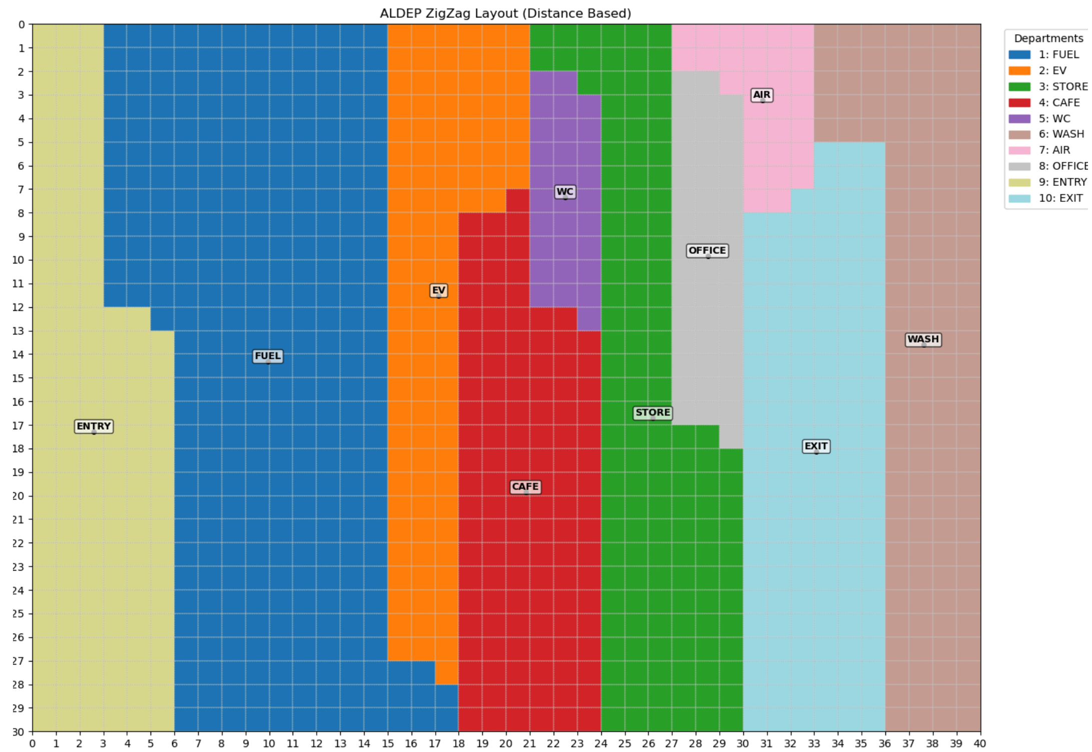

Figure 24. Initial Layout Configuration Derived from ALDEP Logic

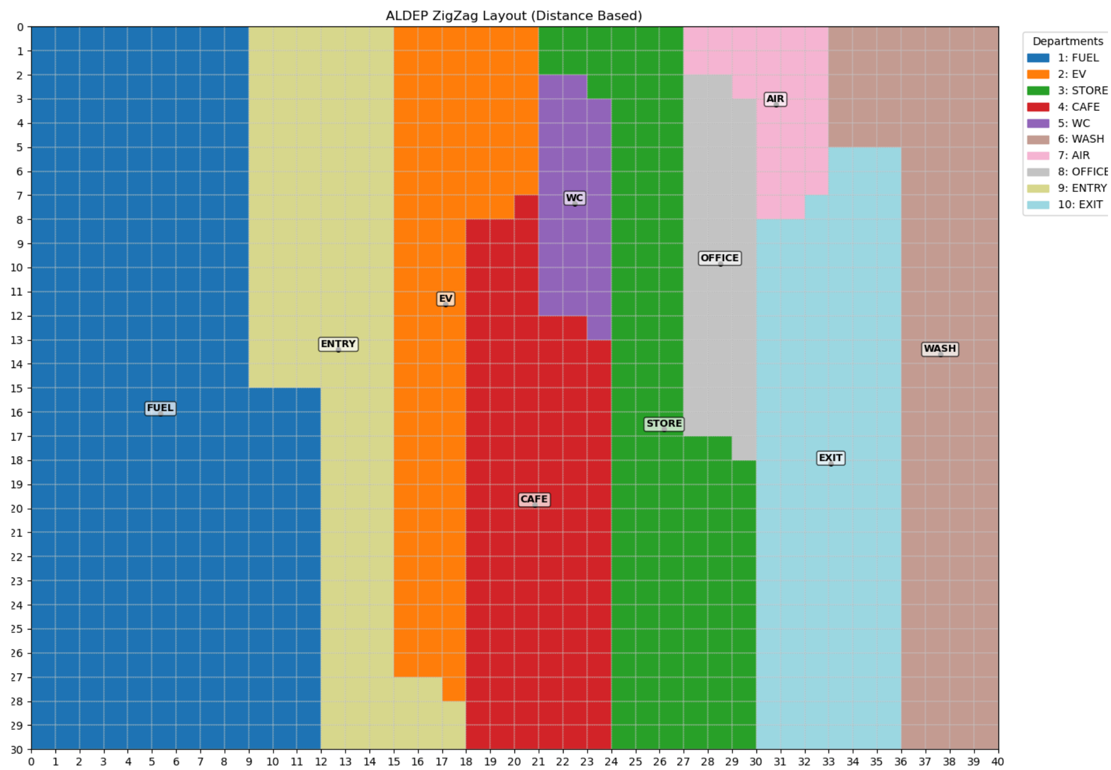

Figure 25. CRAFT Optimization Process - Iteration 1 Sequence Adjustment

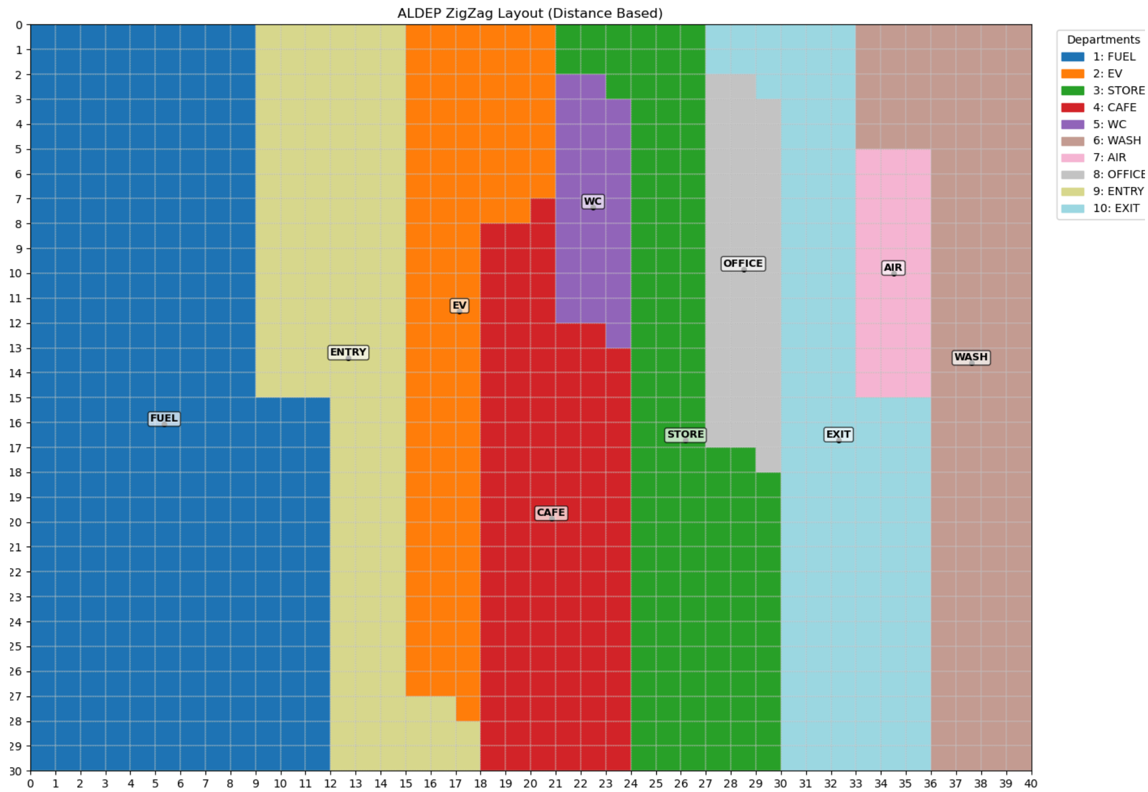

Figure 26. CRAFT Optimization Process - Iteration 2 Sequence Adjustment

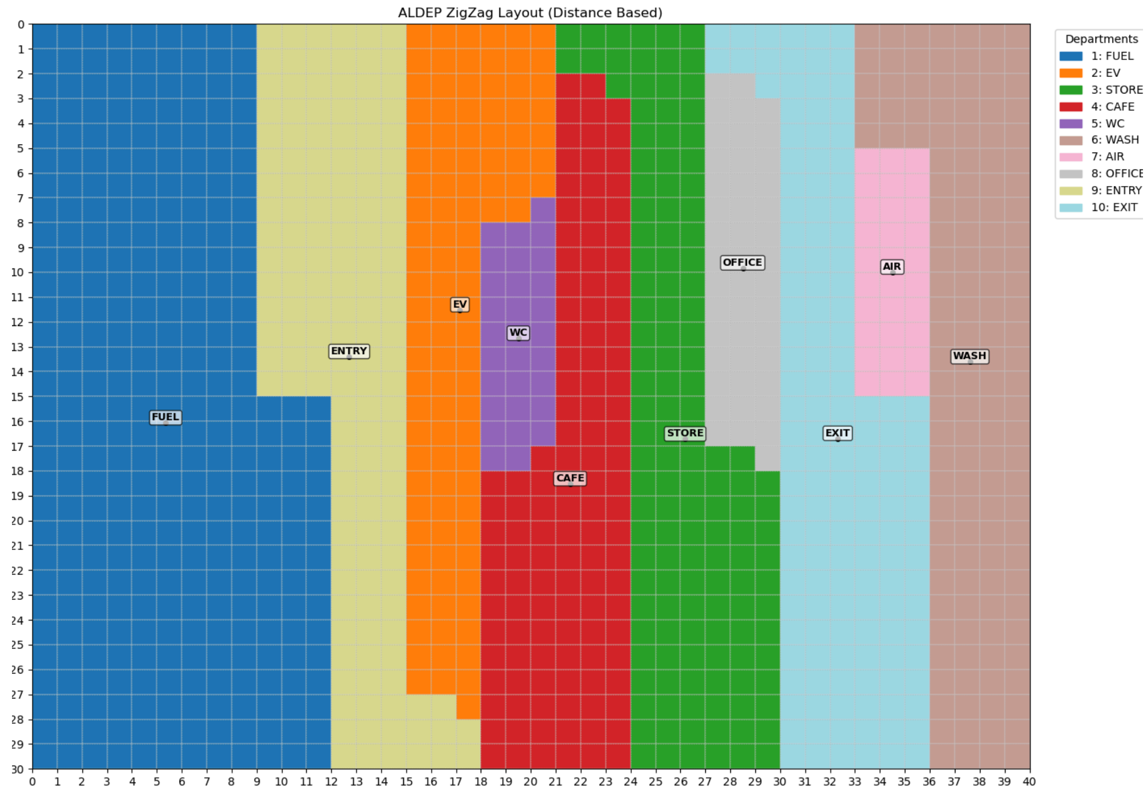

Figure 27. CRAFT Optimization Process - Iteration 3 Sequence Adjustment

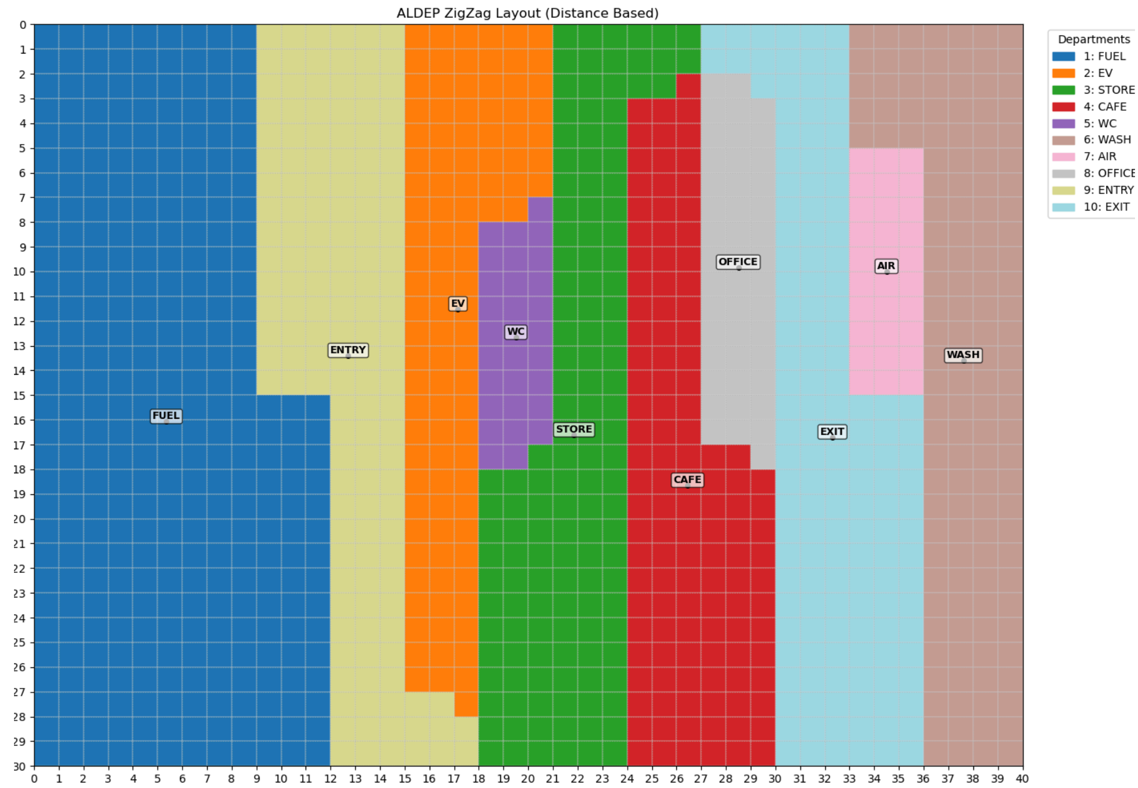

Figure 28. CRAFT Optimization Process - Iteration 4 Sequence Adjustment

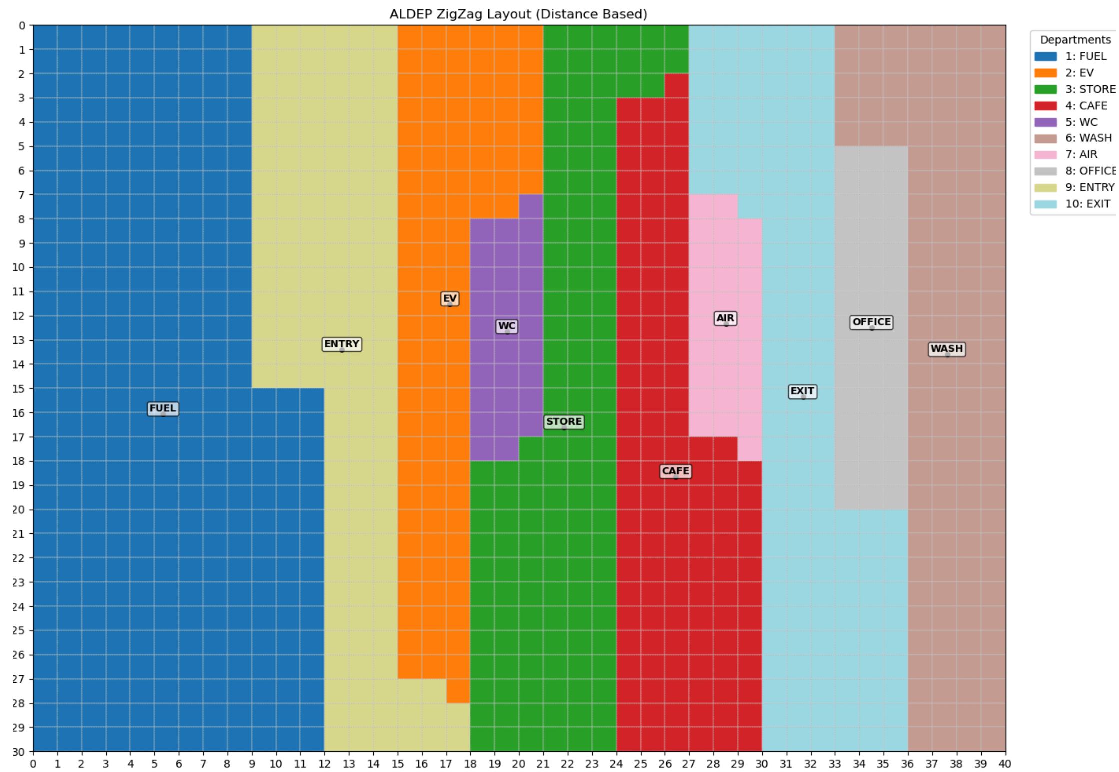

Figure 29. Final Converged Optimal Layout Solution

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